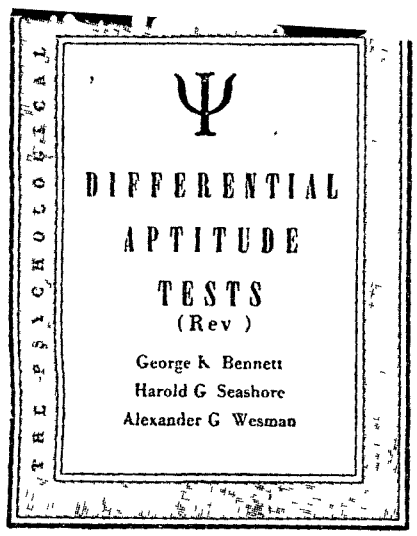


- 246 -
APPENDIX - E,

MECHANICAL REASONING
FORM A

Do not open this booklet until you are told to do so

On your SEPARATE ANSWER SHEET, write your name, address, and other requested information in the proper spaces

In the space after Form, write an A.

Then wait for further instructions.

DO NOT MAKE ANY MARKS IN THIS BOOKLET

MĀNASĀYAN
32 Faiz Bazar
Delhi-7

Modified and reproduced by Special permission of the copyright owner and original publisher THE PSYCHOLOGICAL CORPORATION, NEW YORK NEW YORK All Rights Reserved

Printed in India

Do not make any marks in this booklet

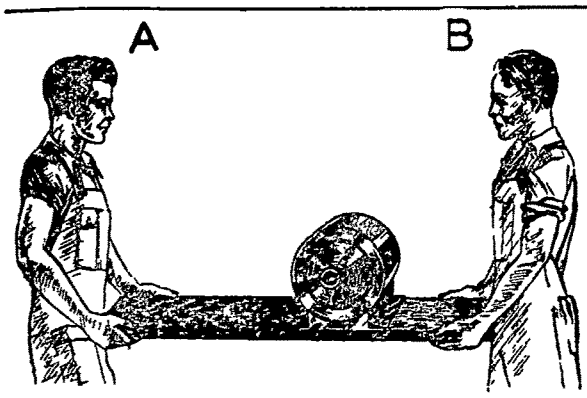
Mark your answers on the separate Answer Sheet

MECHANICAL REASONING

DIRECTIONS

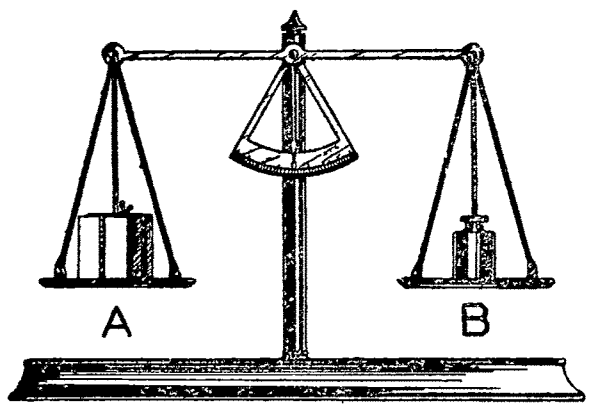
This test consists of a number of pictures and questions about those pictures. Look at Example X on this page to see just what to do. Example X shows a picture of two men carrying a machine part on a board and asks, "Which man has the heavier load? If equal, mark "C." Man "B" has the heavier load because the weight is closer to him than to man "A," so on the separate Answer Sheet you would put a cross (X) in the circle under B, like this → A B C ○ ⊗ ○

Now look at Example Y. The question asks, "Which weighs more? If equal, mark C." As the scale is perfectly balanced, "A" and "B" must weigh the same, so you would put a cross (X) in the circle under C on your separate Answer Sheet, like this → A B C ○ ○ ⊗



X

Which man has the heavier load ?
(If equal, mark C)



Y

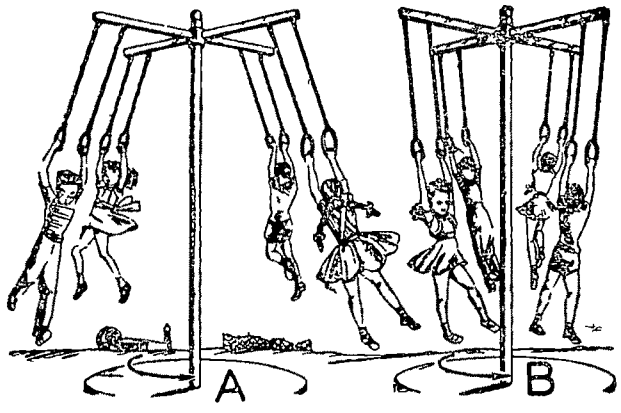
Which weighs more ?
(If equal, mark C.)

On the following pages there are more pictures and questions. Read each question carefully, look at the picture, and mark your answer on the separate Answer Sheet. Do not forget that there is a third choice for every question.

DO NOT TURN OVER THE BOOKLET UNTIL YOU ARE TOLD TO DO SO.

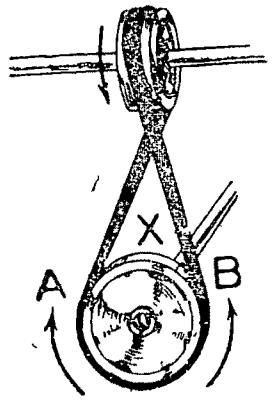
Drawing by Helen Gabryel

PUT YOUR ANSWERS ON THE ANSWER SHEET



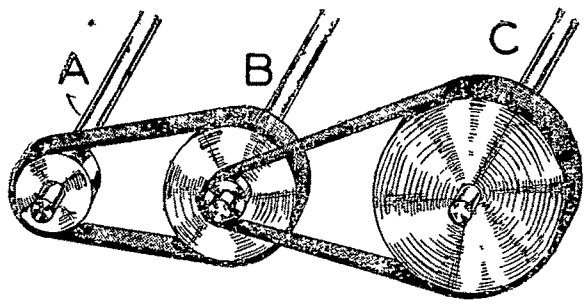
1

In which picture are the children turning faster ?
(If equal, mark C)



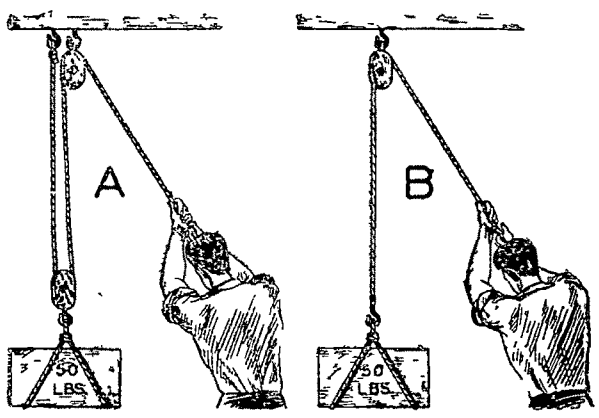
2

When the top pulley turns in the direction shown, which way, A or B, will the lower pulley turn ?
(If it is not possible to know, mark C.)



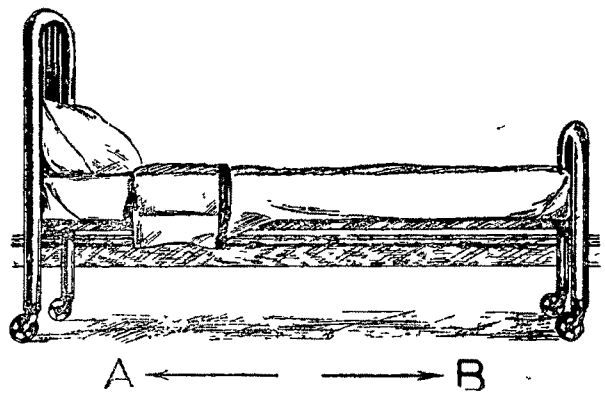
3

Which shaft (rod) will turn most slowly ?



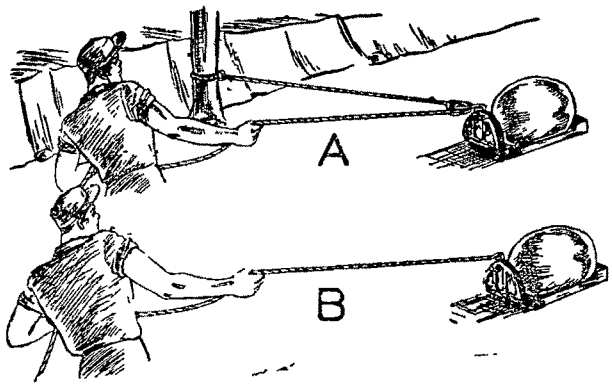
4

Which man must pull harder to lift the weight ?
(If equal, mark C.)



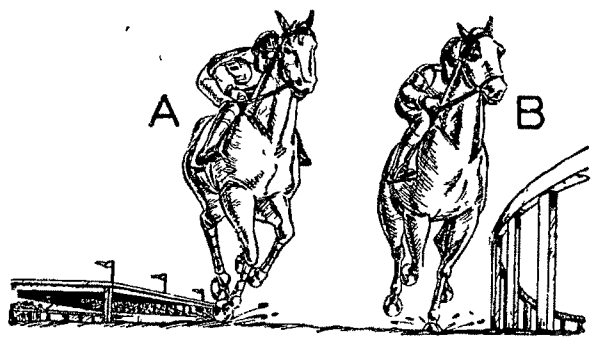
5

In which direction, A or B, was this bed just moved ?
(If it is not possible to know, mark C.)



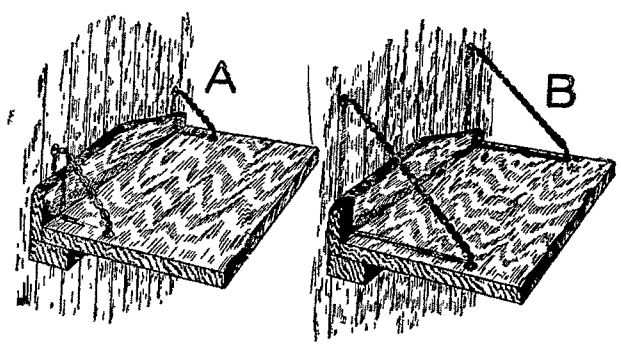
6

Which man has to pull harder ?
(If equal, mark C.)



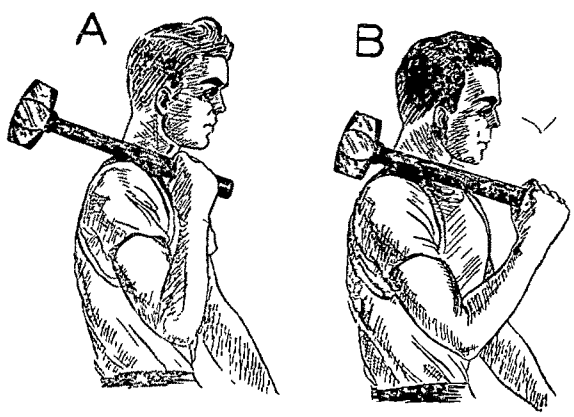
7

Which horse must go faster to hold his place on the turn ?
(If equal, mark C.)



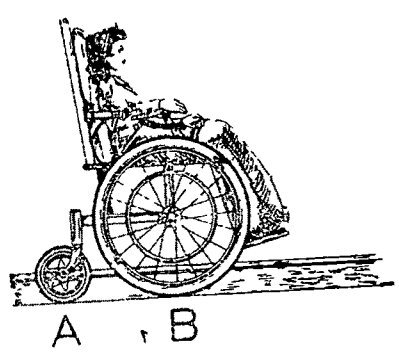
8

Which shelf is stronger ?
(If equal, mark C.)



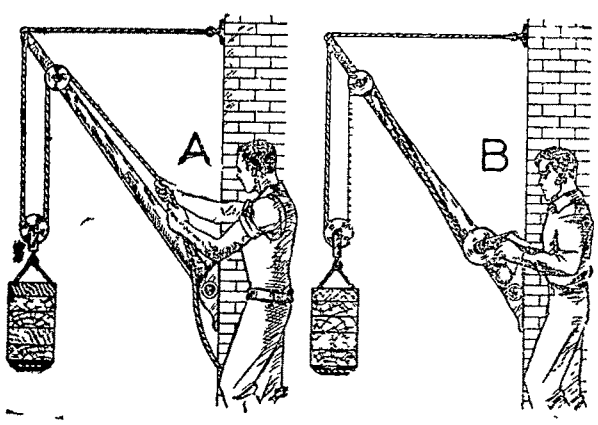
9

Which is the harder way to carry the hammer ?
(If equal, mark C)



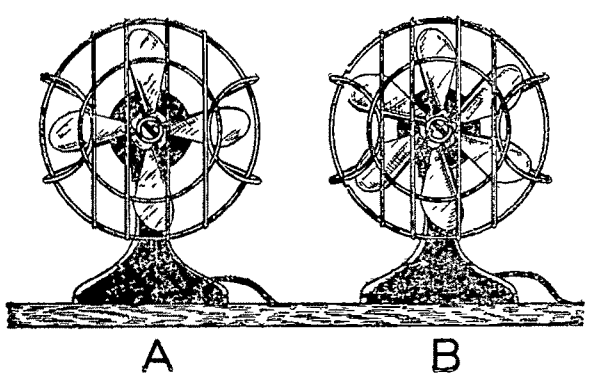
10

Which wheel will turn faster ?
(If equal, mark C.)



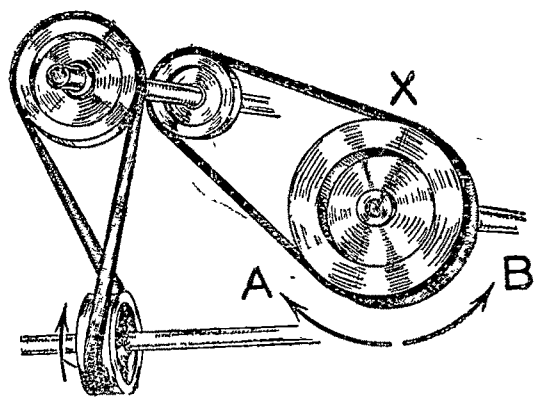
11

Which man can lift the weight more easily ?
(If equal, mark C.)



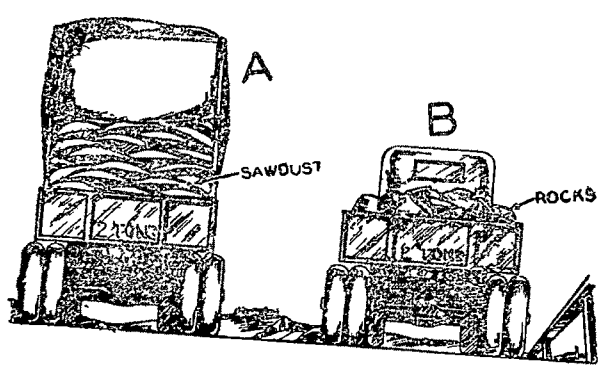
12

Which fan needs the more powerful motor ?
(If equal, mark C)



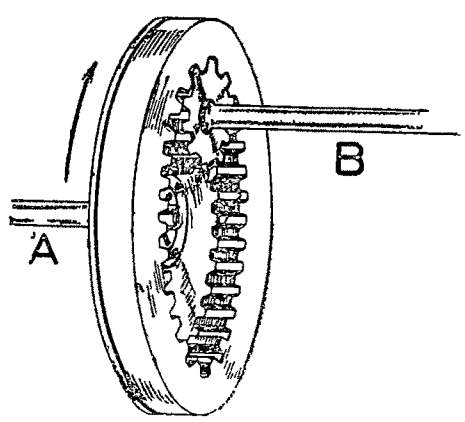
13

Which way, A or B, will pulley "X" turn ?
(If it is not possible to know, mark C.)



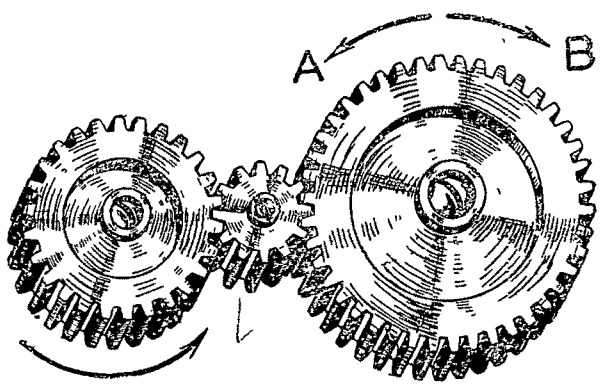
14

Which truck will turn over more easily ?
(If equal, mark C.)



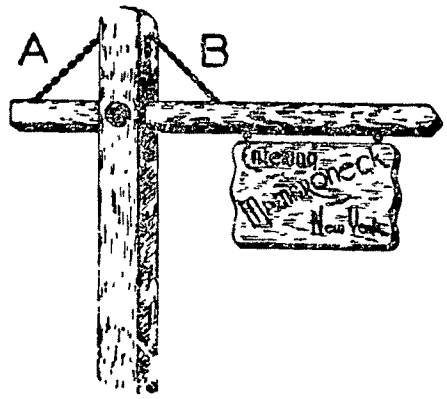
15

Which shaft (rod) turns faster ?
(If equal, mark C.)



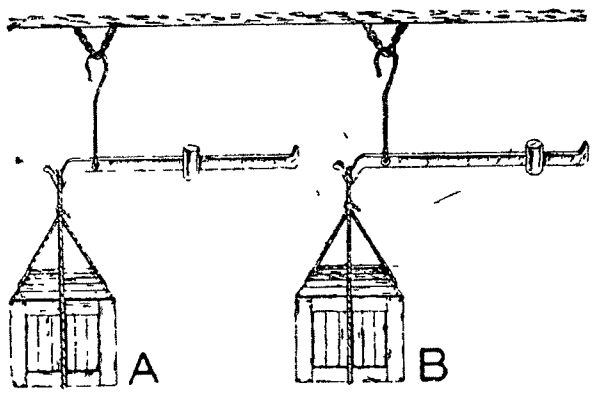
16

When the small gear on the left turns in the direction shown, which way will the big gear "X" turn ?
(If it is not possible to know, mark C.)



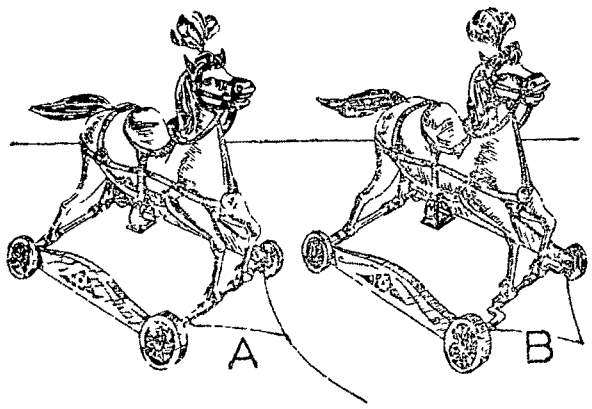
17

Which chain alone will hold up this sign ?
(If either alone could hold up the sign, mark C.)



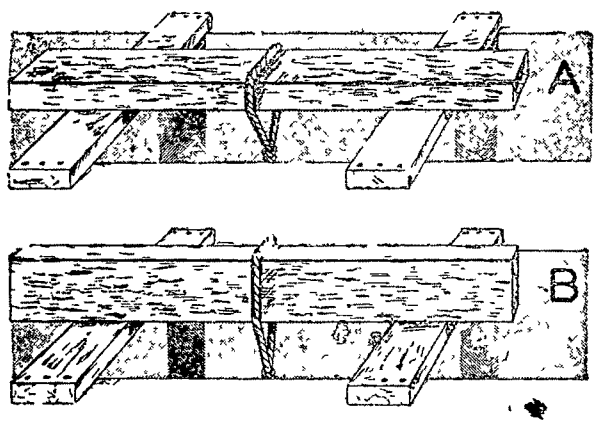
18

Which box weighs more ?
(If equal, mark C.)



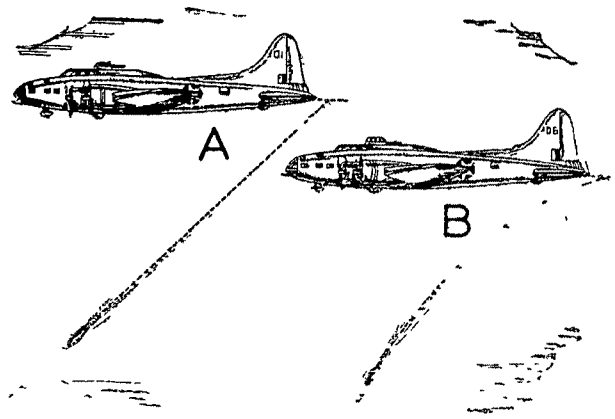
19

Which horse will jump more when it is pulled ?
(If equal, mark C.)



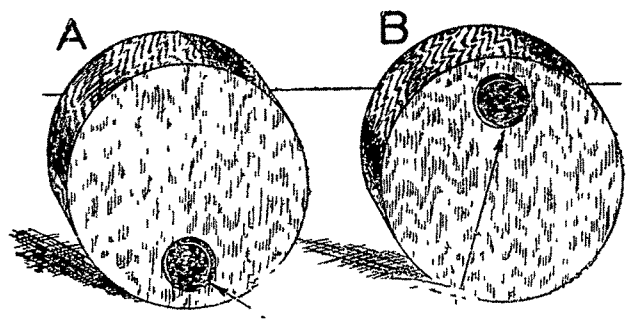
20

In which picture can you safely put a heavier weight on the rope ?
(If equal, mark C.)



21

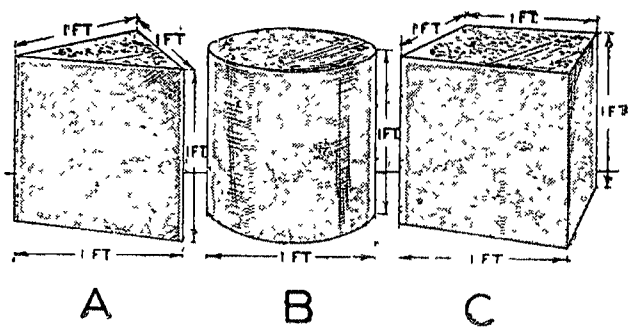
Which drawing shows how a bomb really falls ?
(If both, mark C.)



22

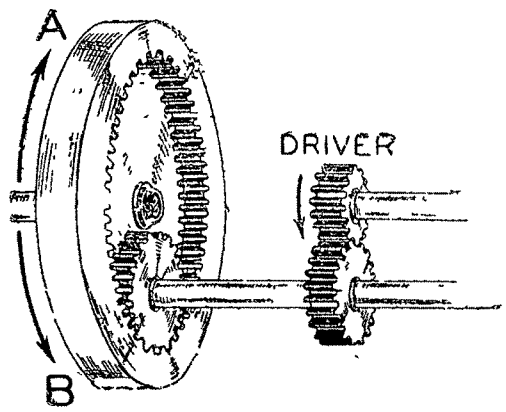
Which picture shows how this wooden circle will stand ?
(If neither, mark C.)

HOLES FILLED WITH IRON



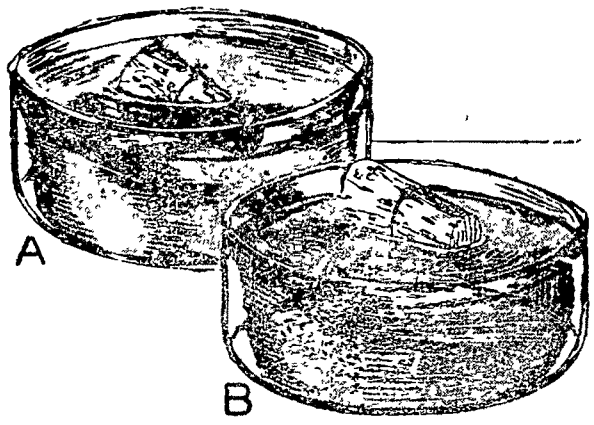
23

Which weighs least ?



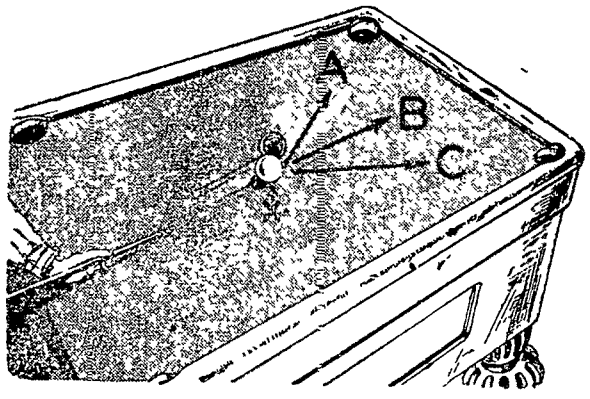
24

When the driver turns in the direction shown, which way, A or B, will the big left-hand gear turn ?
(If it is not possible to know, mark C.)



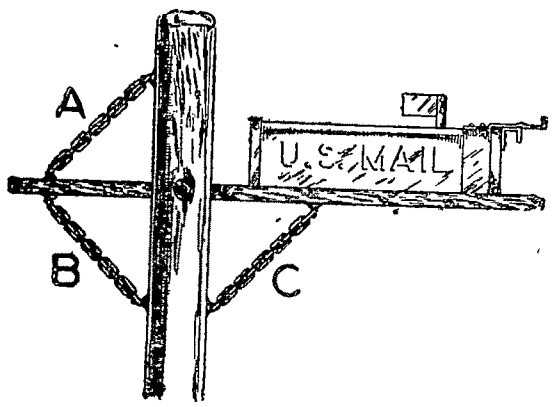
25

Which liquid is heavier ?
(If equal, mark C.)



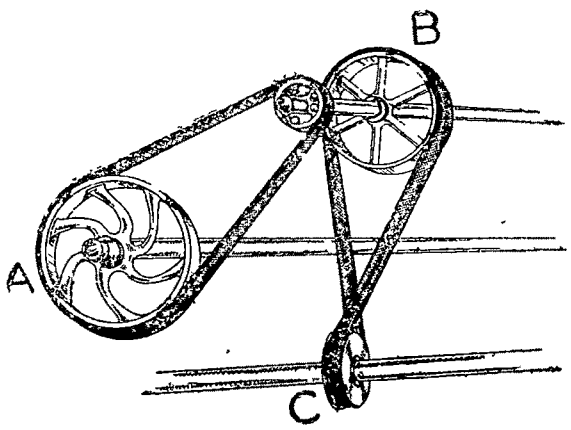
26

After hitting the black ball, which way will ball "X" go ?



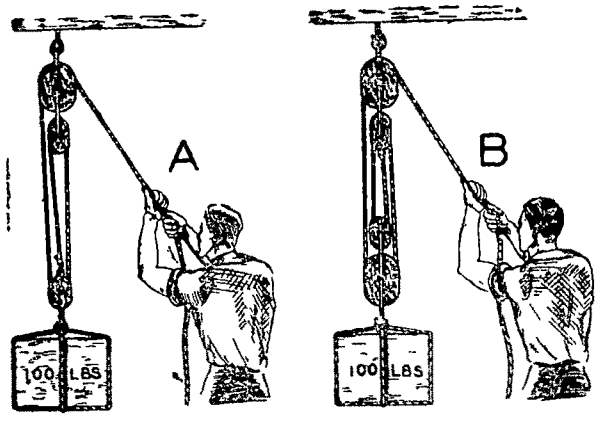
27

Which one piece of chain is needed to support the mail box ?



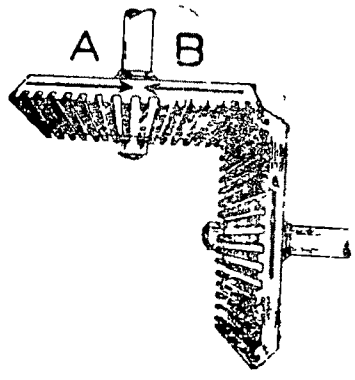
28

Which shaft (rod) turns fastest ?



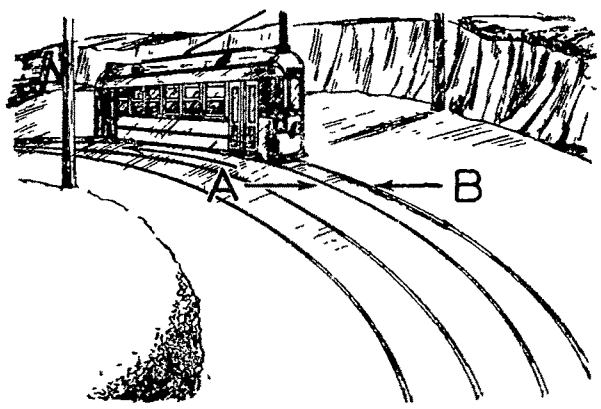
29

Which man can lift the load more easily ?
(If equal, mark C.)



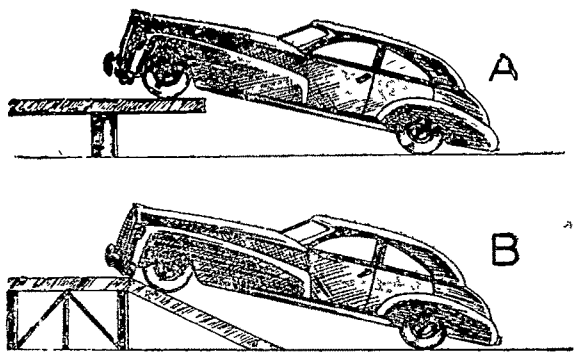
30

When the right hand gear turns in the direction shown, which way does the top gear turn ?
(If neither, mark C.)



31

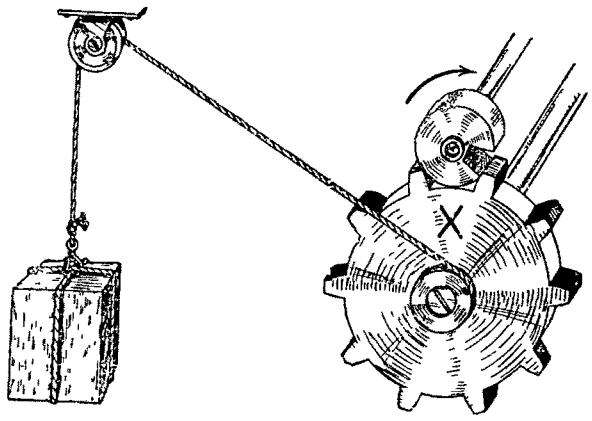
Which rail should be higher ?
(If equal, mark C.)



32

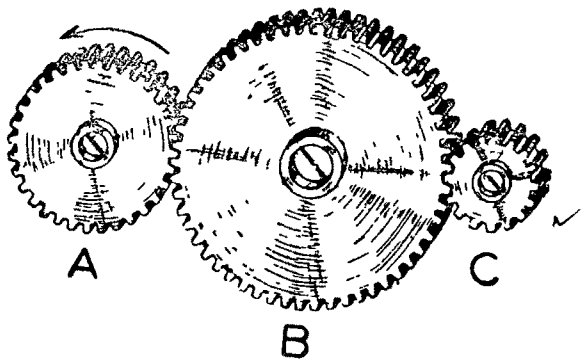
Which car is less likely to roll backwards ?
(If equal, mark C.)

PUT YOUR ANSWERS ON THE ANSWER SHEET



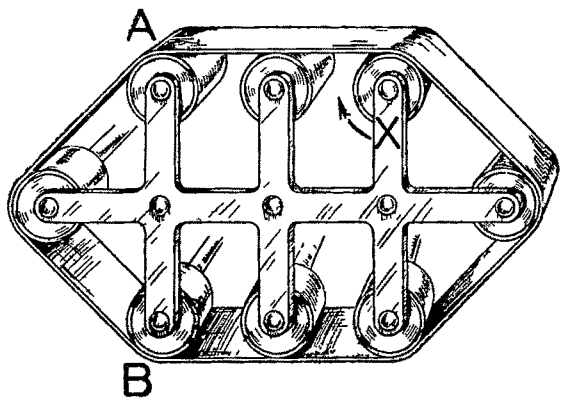
33

The top of wheel "X" will move:
(A) steadly (regularly) to the right;
(B) steadly (regularly) to the left;
(C) by jerks (starting and stopping) to the left.



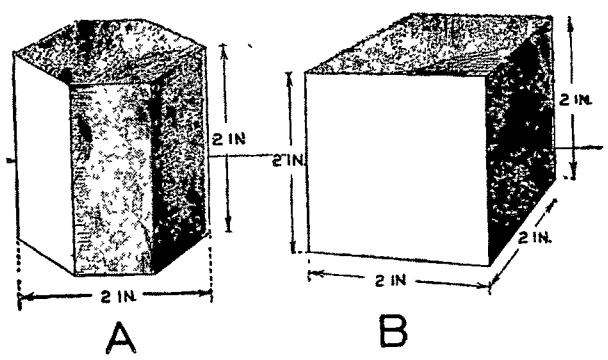
34

Which gear turns most in a minute?



35

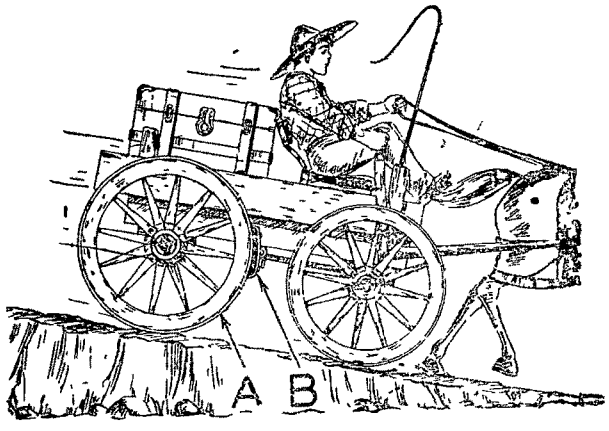
Which roller turns the same way as the roller at "X" ?
(If both, mark C.)



36

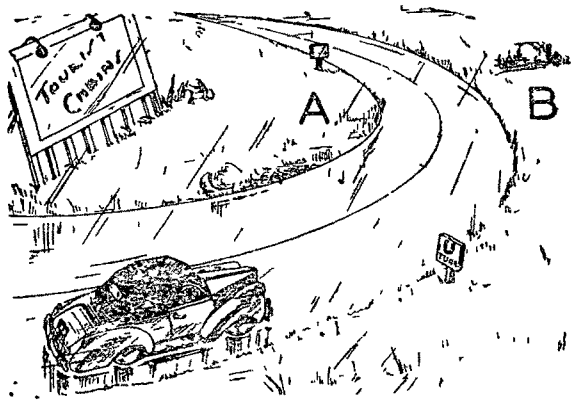
Which weighs more ?
(If equal, mark C.)

Do Not Stop. Go On to the Next Page.



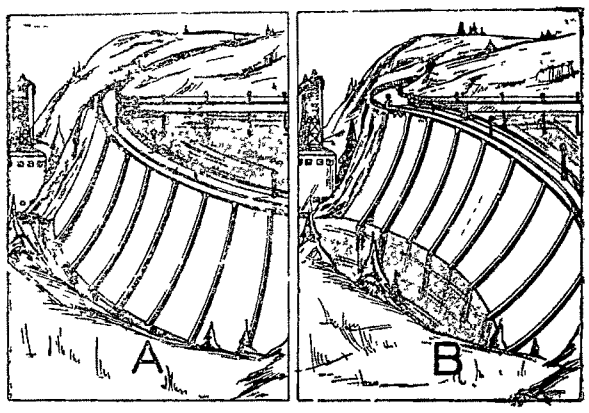
37

When the brake is put on which part gets hotter ?
(If equal, mark C.)



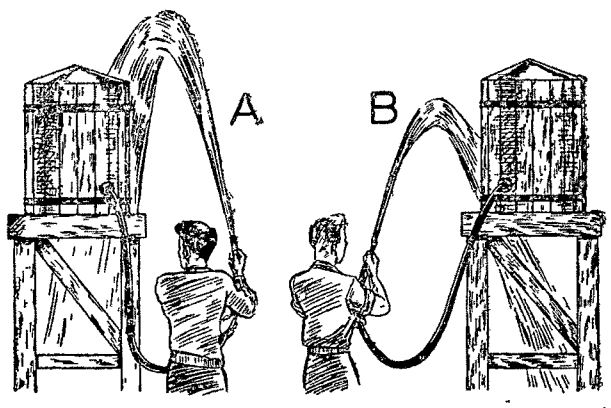
38

Towards which side of the road is the car more likely to skid (slide) ?
(If equal, mark C.)



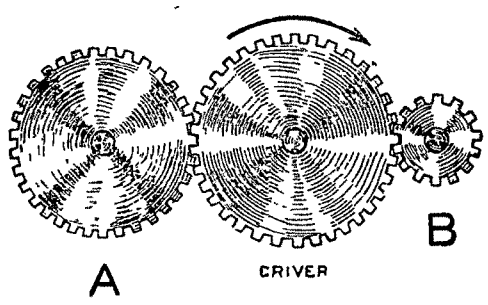
39

Which dam is stronger ?
(If equal, mark C.)



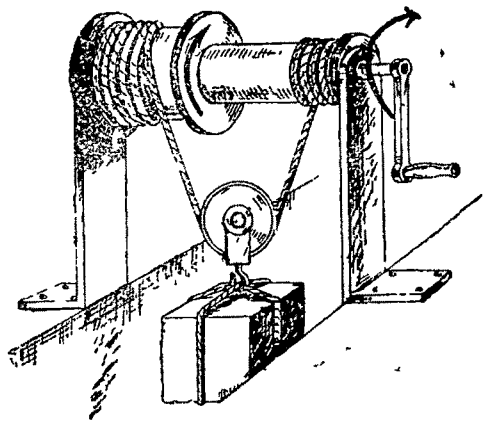
40

Which picture is correct ?
(If both, mark C.)



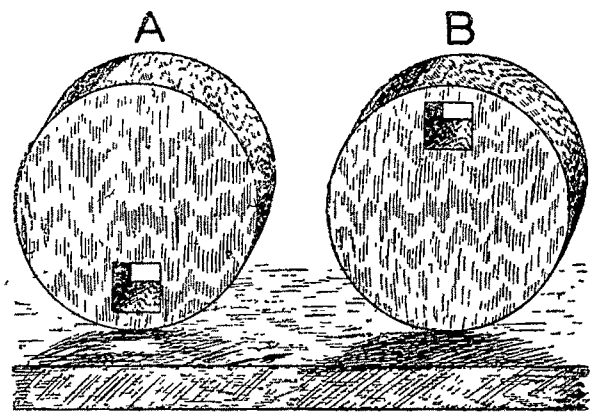
41

Which gear turns the same way as the driver ?
(If neither, mark C.)



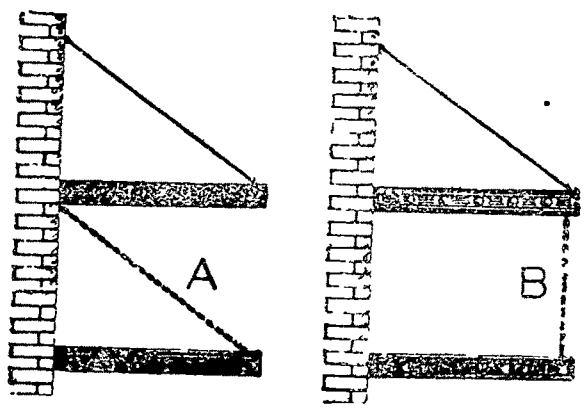
42

When the crank is turned in the direction shown, the weight will :
(A) fall;
(B) stand still;
(C) rise.



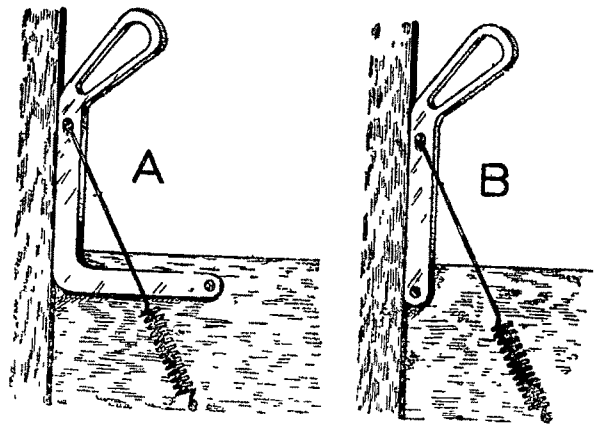
43

Which picture shows how this wooden circle will stand ?
(If neither, mark C.)



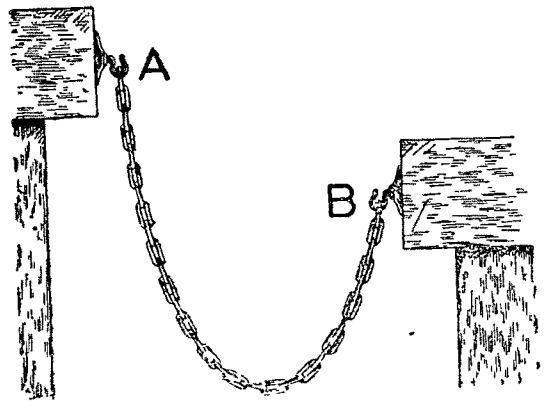
44

Which chain has more strain upon it ?
(If equal, mark C.)



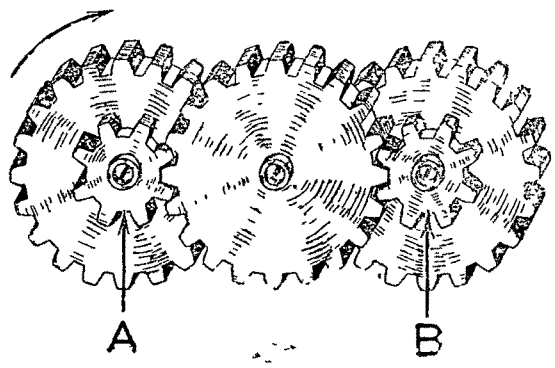
45

In which picture will the spring hold the handle where it now is ? (If both, mark C.)



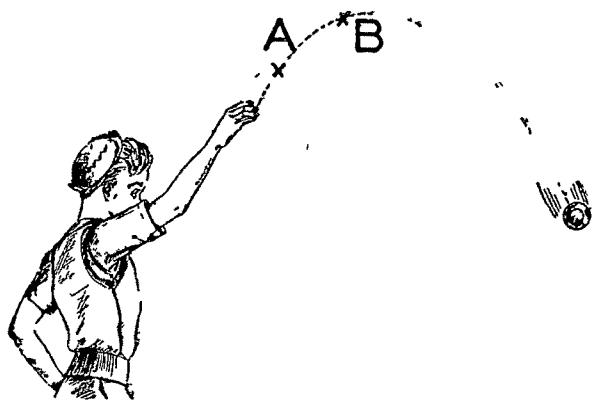
46

Which hook supports more weight ? (If equal, mark C)



47

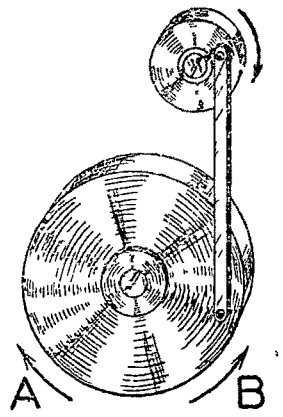
Which gear turns slower ? (If equal, mark C.)



48

At which point was the ball going faster ?

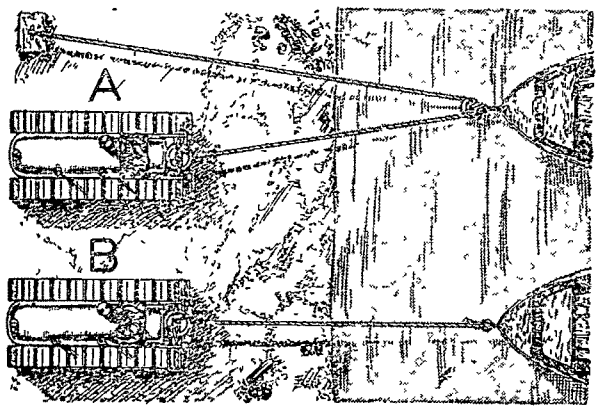
(If equal, mark C.)



49

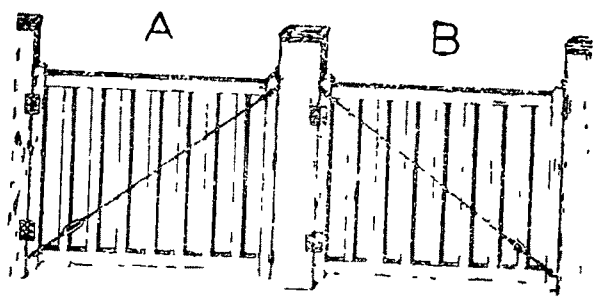
When the small wheel is turned around, the big wheel will :

(A) turn in direction A;
(B) turn in direction B;
(C) move back and forth



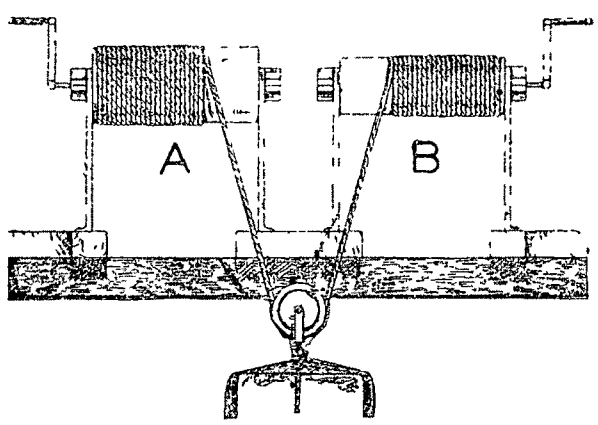
50

Which tractor must go further to pull the boat up on the beach ?
(If equal, mark C.)



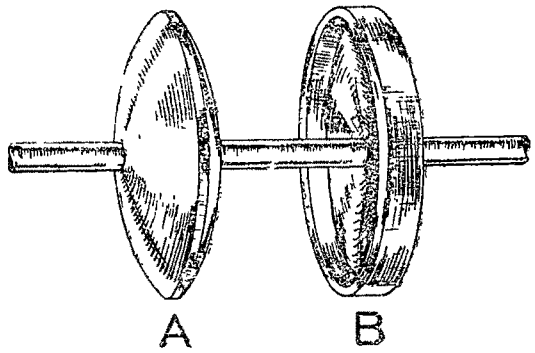
51

Which gate is better braced (made stronger) by the wire across it ?
(If equal, mark C.)



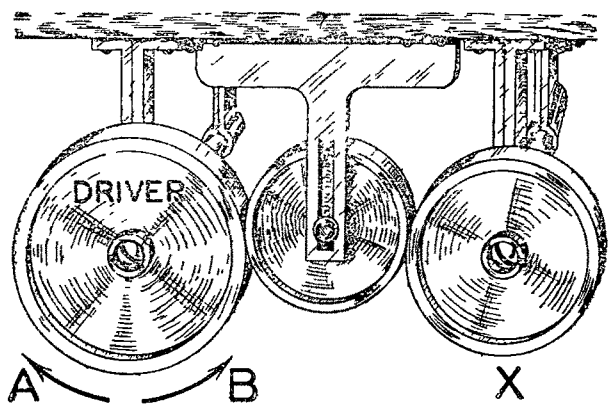
52

Which windlass (crank) will be harder to turn in order to lift the weight ?
(If equal, mark C.)



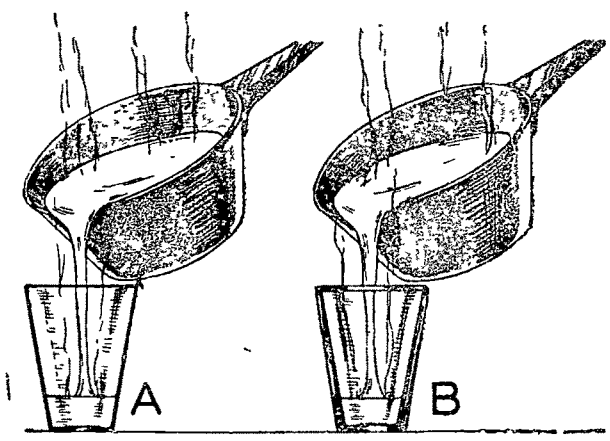
53

Which wheel is safer (less likely to break) when spun (turned) at high speed ?
(If equal, mark C.)



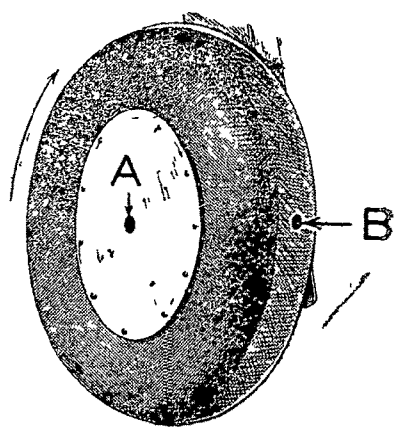
54

Which way must the driver turn to drive the wheel "X" ?
(If it can drive "X" by turning in either direction, mark C.)



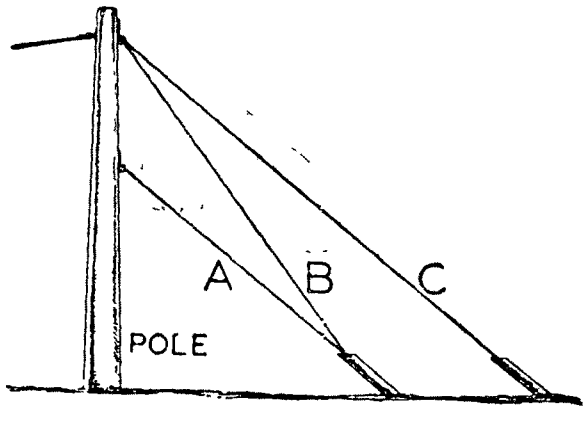
55

Which glass is more likely to break ?
(If equal, mark C.)



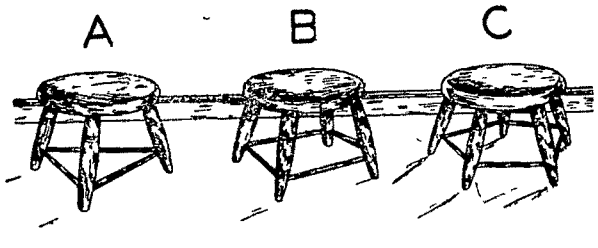
56

Which point moves faster when the wheel turns ?
(If equal, mark C.)



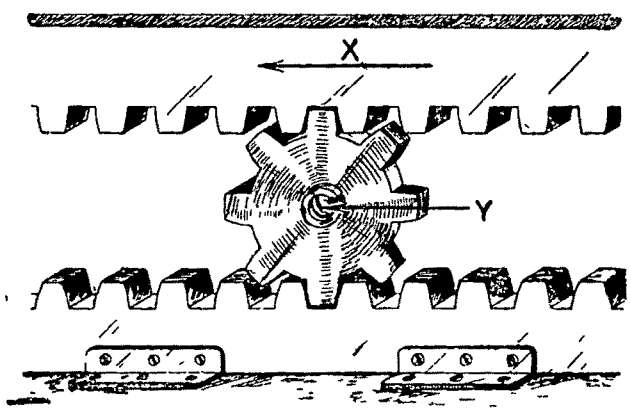
57

Which one of the three pieces of wire will give this pole the best support ?



58

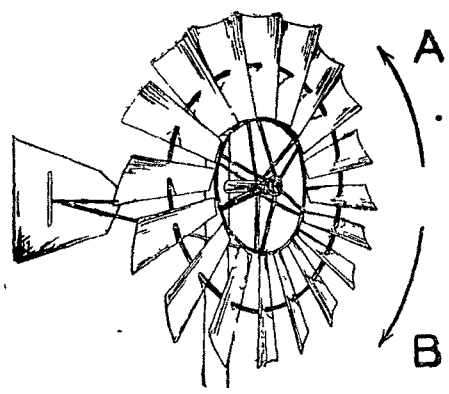
Which stool will be steadiest on uneven ground ?



59

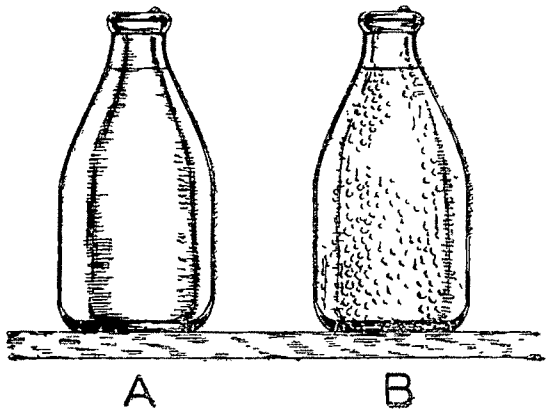
If "X" moves two feet in the direction shown, the centre of the gear "Y" will move :

(A) more than two feet;
(B) less than two feet;
(C) two feet.



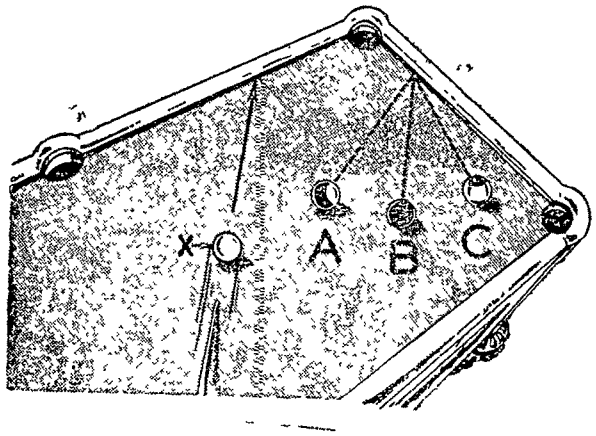
60

In which direction is this wind-mill more likely to turn when the wind blows ?
(If it is not possible to know, mark C.)



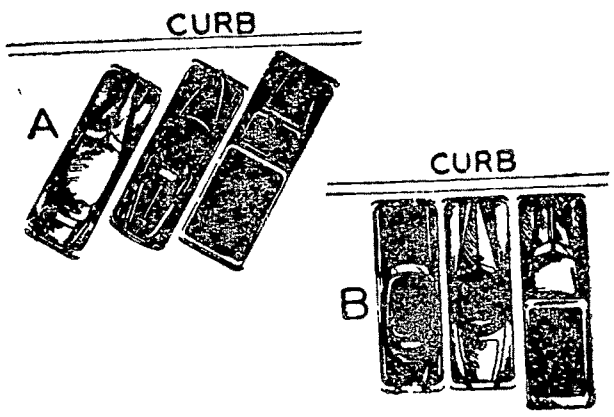
61

Which bottle has just been taken from a refrigerator or ice-box ?
(If neither, mark C.)



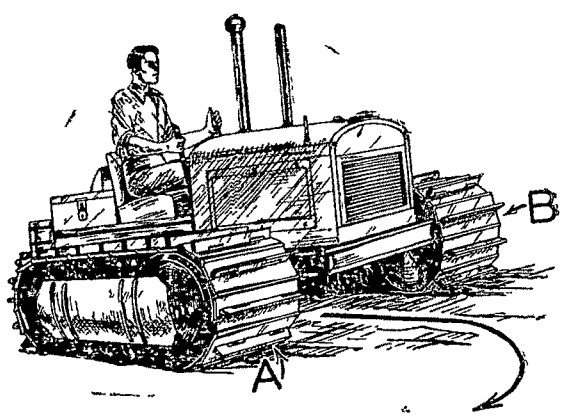
62

Which of these balls will the white ball "X" hit ?



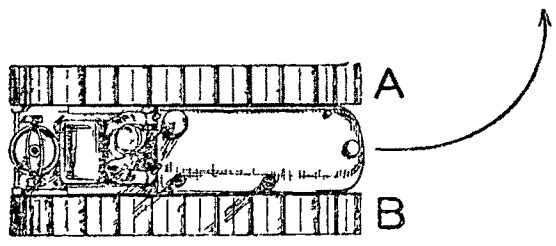
63

Which way can more cars be parked in a block ?
(If equal, mark C.)



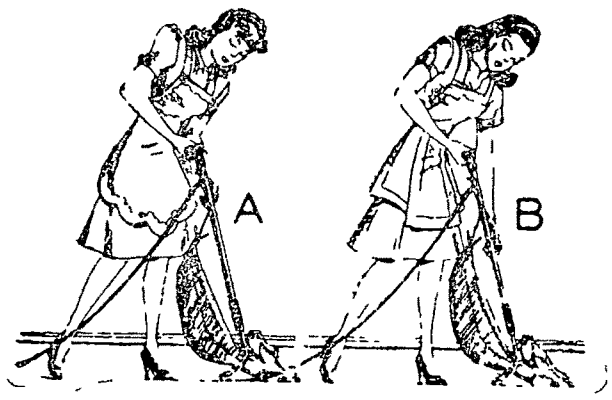
64

Which tread, A or B, should move more slowly for the tractor to turn in the direction shown ?
(If neither, mark C.)



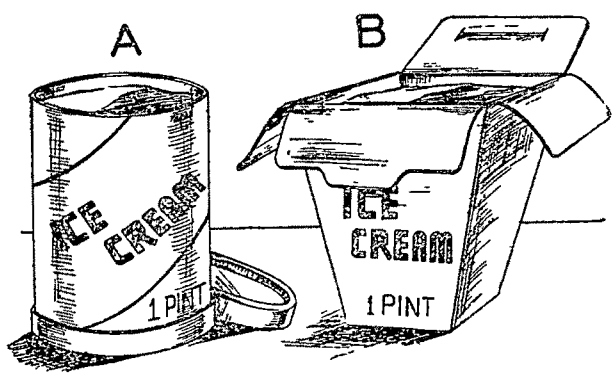
65

Which tread A or B, should be run faster to move (turn) the tractor in the direction shown ?
(If neither, mark C.)



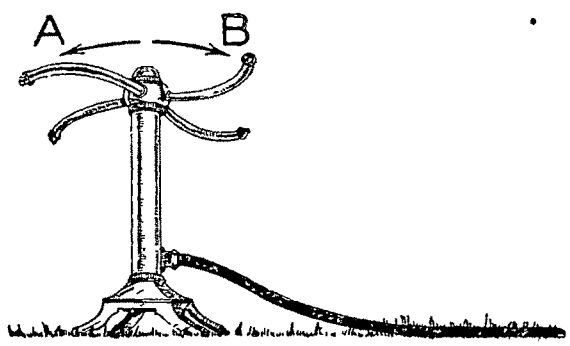
66

Which girl can lift the cleaner more easily ?
(If equal, mark C.)



67

In which card board box will the icecream stay hard longer ?
(If equal, mark C.)



68

When the water is turned on, which way will the sprinkler turn ?
(If it is not possible to know, mark C)

NAME Appendix E2 DATE

PLACE (SCHOOL) CITY

AGE SEX CLASS

YEARS MONTHS MALE OR FEMALE

RAW SCORE	PERCENTILE
NORMS USED	

265 DIFFERENTIAL APTITUDE TESTS

MECHANICAL REASONING

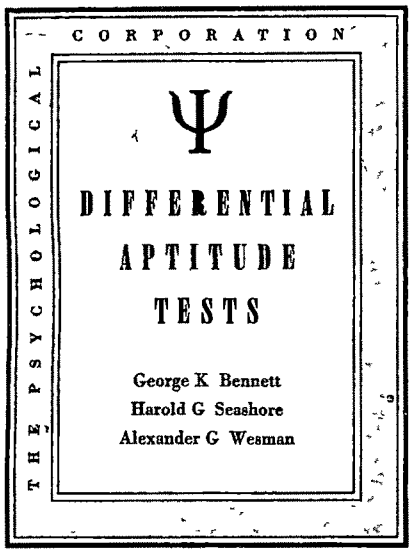
ANSWER SHEET

FORM

ERASE COMPLETELY ANY ANSWER YOU WISH TO CHANGE.
BE SURE YOUR MARK IS INSIDE THE CIRCLE.

| | | |

APPENDIX - F1



SPACE RELATIONS

FORM A

Do not open this booklet until you are told to do so

On your SEPARATE ANSWER SHEET, write your name, address, and other requested information in the proper spaces

In the space after **Form,** write **an A.**

Then wait for further instructions

DO NOT MAKE ANY MARKS IN THIS BOOKLET

MĀNASĀYAN
32 Netaji Subhash Marg
New Delhi-110002

Modified and reproduced by Special permission of the original publisher and copyright owner, THE PSYCHOLOGICAL CORPORATION, NEW YORK, NEW YORK All Rights Reserved

Printed in India.

Do not make any marks in this booklet

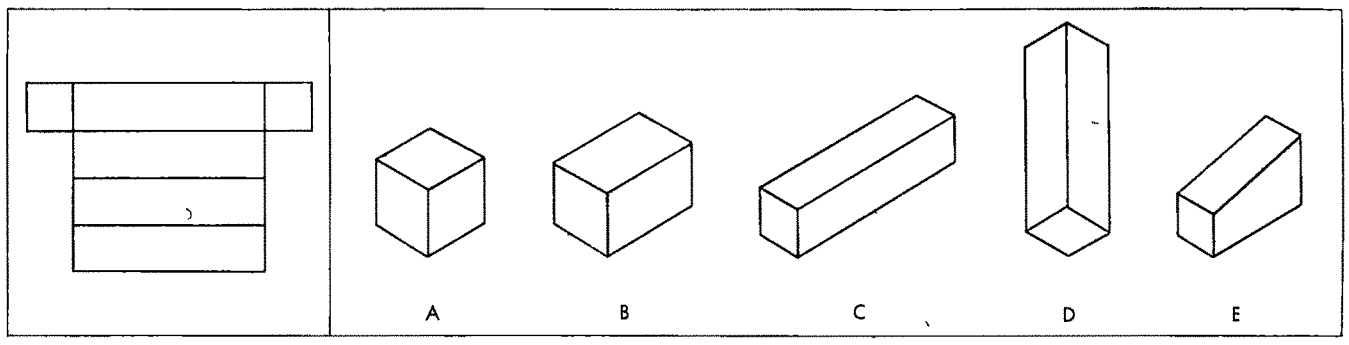
Mark your answers on the separate Answer Sheet

SPACE RELATIONS

DIRECTIONS

This test consists of forty patterns which can be folded into figures. For each pattern, five figures are shown. You are to decide which of these figures can be made from the pattern shown. The pattern always shows the outside of the figure Here is an example·

EXAMPLE X



Which of these five figures — A, B, C, D, E — can be made from the pattern in Example X? A and B certainly cannot be made, they are not the right shape C and D are correct both in shape and size You cannot make E from this pattern

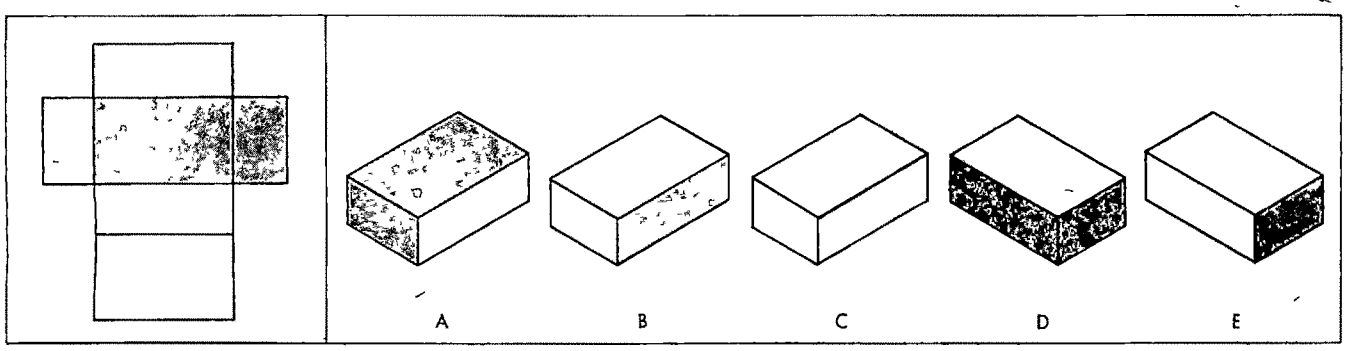
— In the test there will always be a row of five figures for each pattern

— In every row there is at least one correct figure

— Usually more than one is correct In fact, in some cases, all five may be correct

Now look at the pattern for Example Y and the five choices for it Note that when the pattern is folded, the figure must have two gray surfaces One of these is a large surface which could be either the top or bottom of a box The other is a small surface which would be one end of the box

EXAMPLE Y



Notice — all the 'boxes" made from this pattern are correct in **shape,** but the sides which you see are different Some of these figures can be made from this pattern while others cannot Let us look at them

— Figure A is correct If the large gray surface is shown as the top, ther the end surface of gray can be shown facing towards you

— Figure B is wrong The **long, narrow** side is not gray in the pattern

— Figure C is correct The two gray surfaces can both be hidden by placing **the** large gray surface at the bottom and the gray end to the back

— Figure D is wrong The gray end is all right, but there is no long gray side in the pattern

— Figure E is correct One can show the box so that the large gray surface is at the bottom (as it was in C), but with the gray end showing at the front

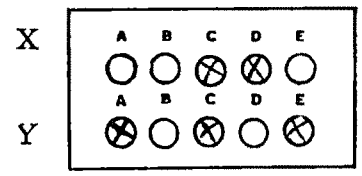
So, you see, there are three figures (A, C and E) which can be made from the pattern in Example Y, and two figures (B and D) which cannot be made from this pattern.

Remember that the surface you see in the pattern must always be the OUTSIDE surface of the completed figure

Now let's see how we mark our answers on the separate Answer Sheet A sample is shown here

For Example X we found that only figures C and D could be made, so the circles under C and D opposite X are marked X. For Example Y, A is a correct figure, C is correct, and E is correct, so opposite Y we have marked X's in the circles under A, C and E.

SAMPLE OF ANSWER SHEET



In taking the test

— Study each pattern

— Decide which of the figures can be made from the pattern

— Show your choices on the Answer Sheet by blackening in the little space under the letter which is the same as that of the figure you have chosen in the booklet

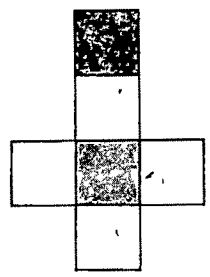
— If you decide a certain figure cannot be made from the Pattern, make no mark on the Answer Sheet

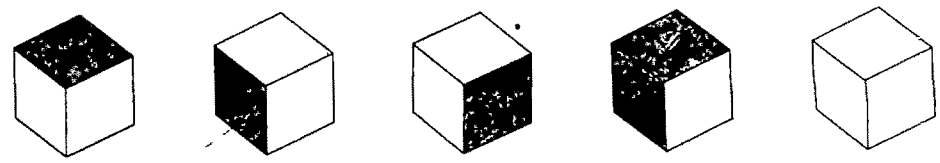
Do Not Write Anything in This Booklet

Use Separate Answer Sheet

DO NOT TURN THE PAGE UNTIL YOU ARE TOLD TO DO SO.

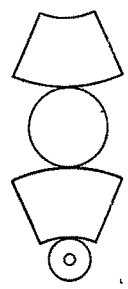
1

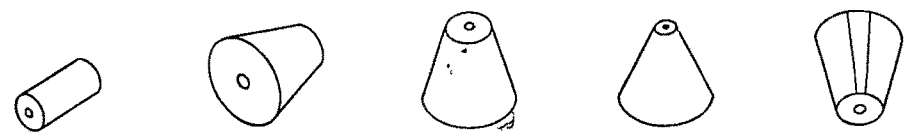




A B C D E

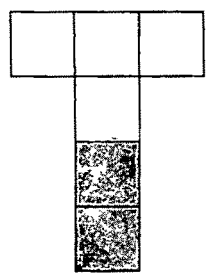
2

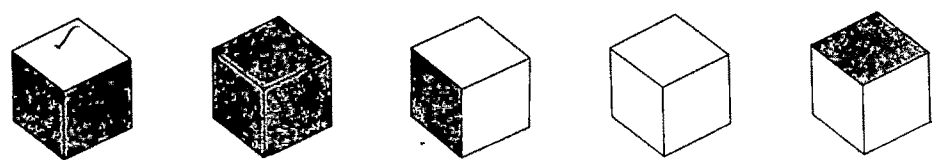




A B C D E

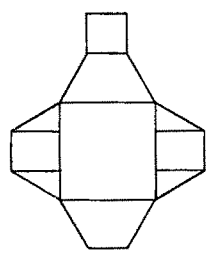
3

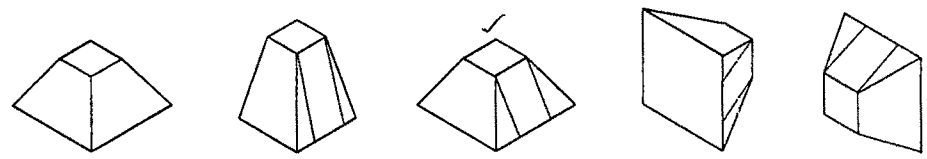




A B C D E

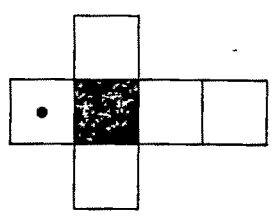
4

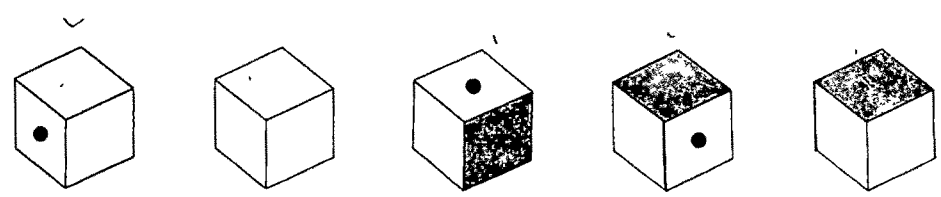




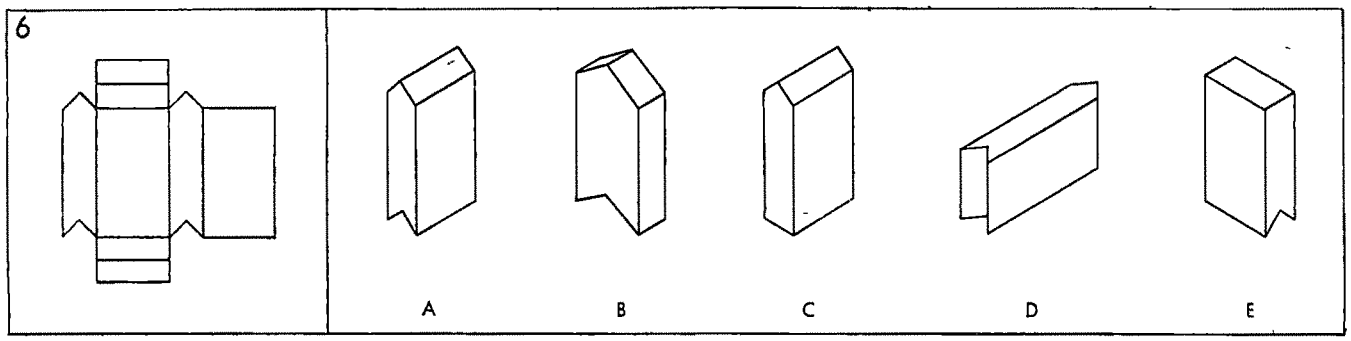
A B C D E

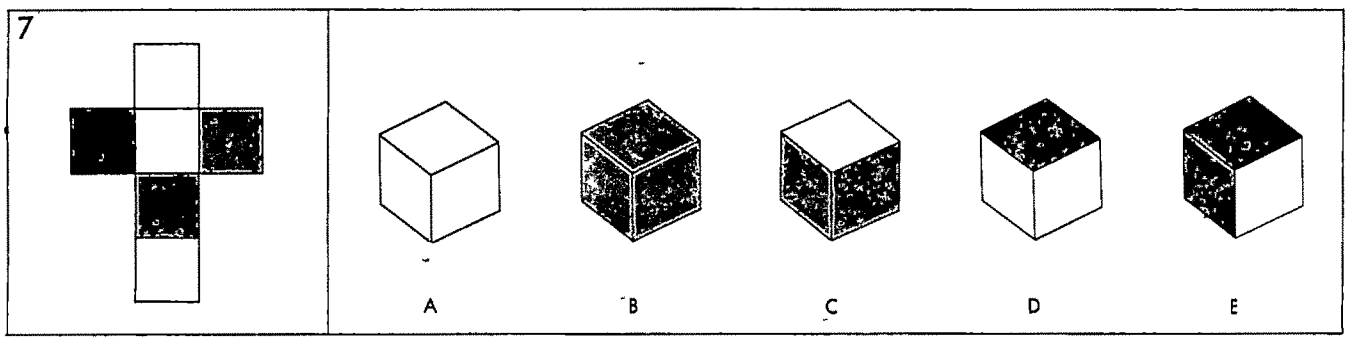
5

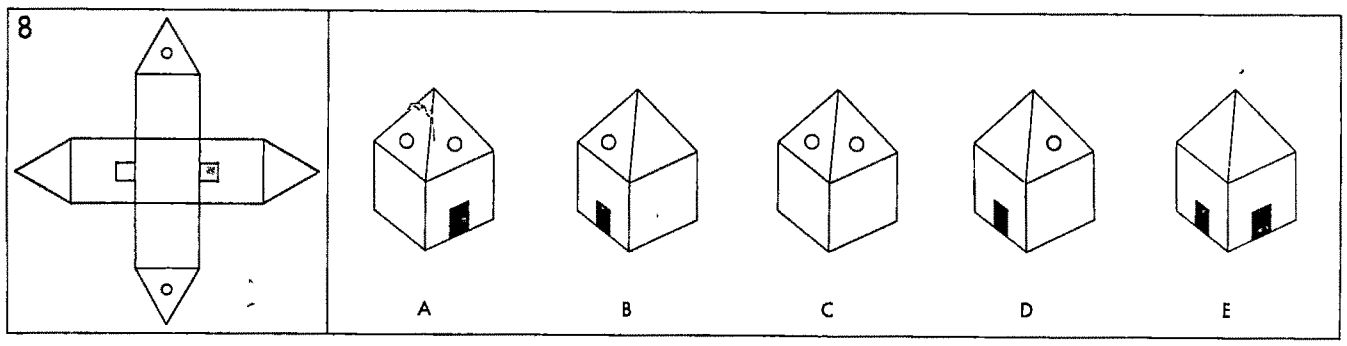


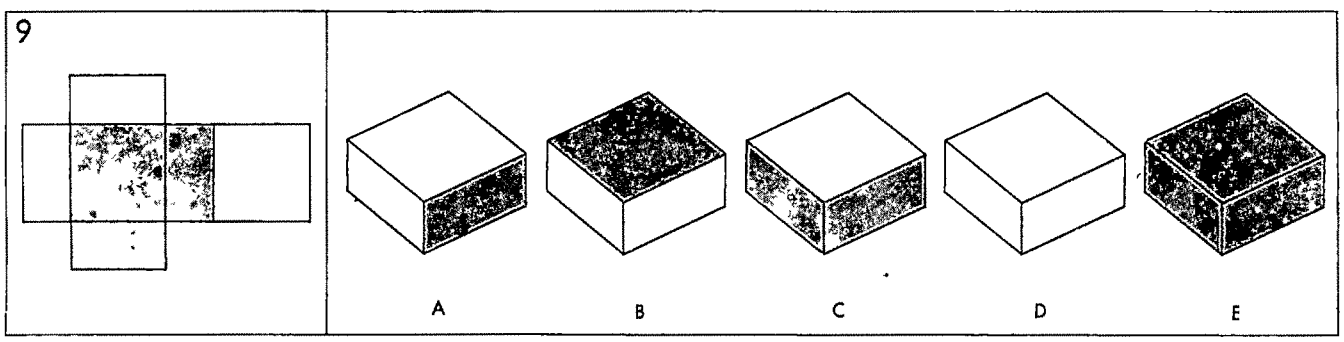


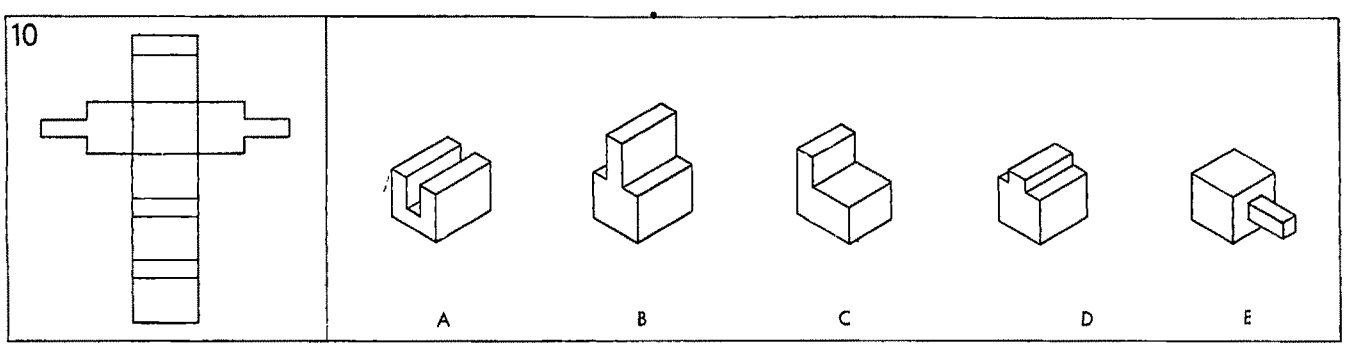
A B C D E


6
A
B
C
D
E

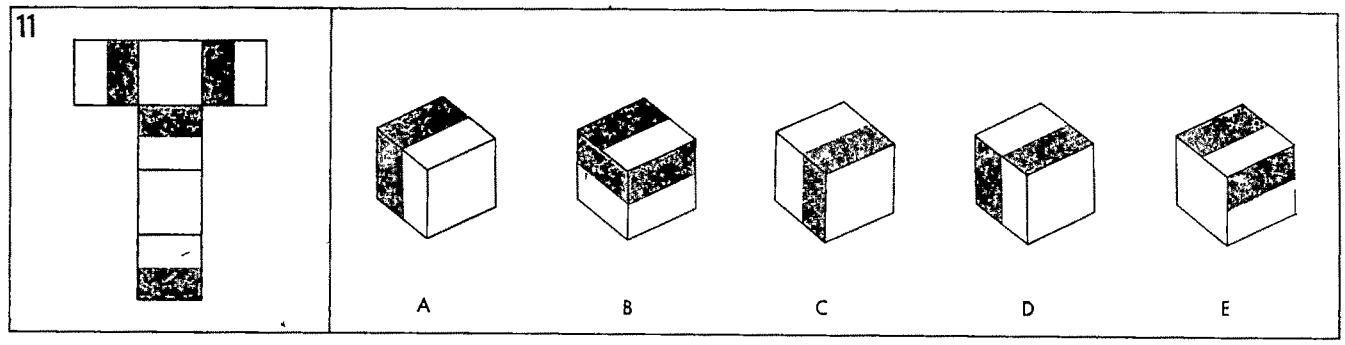

7
A
B
C
D
E


8
A
B
C
D
E

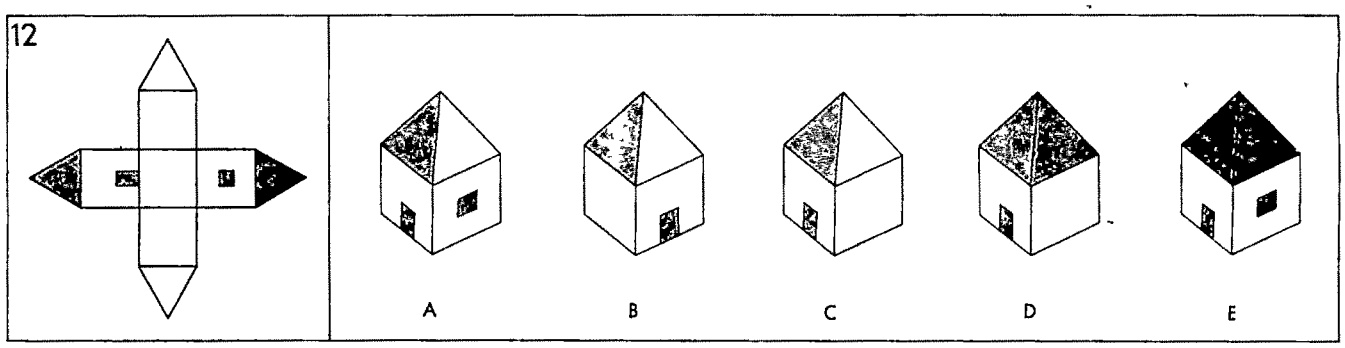

9
A
B
C
D
E


10
A
B
C
D
E

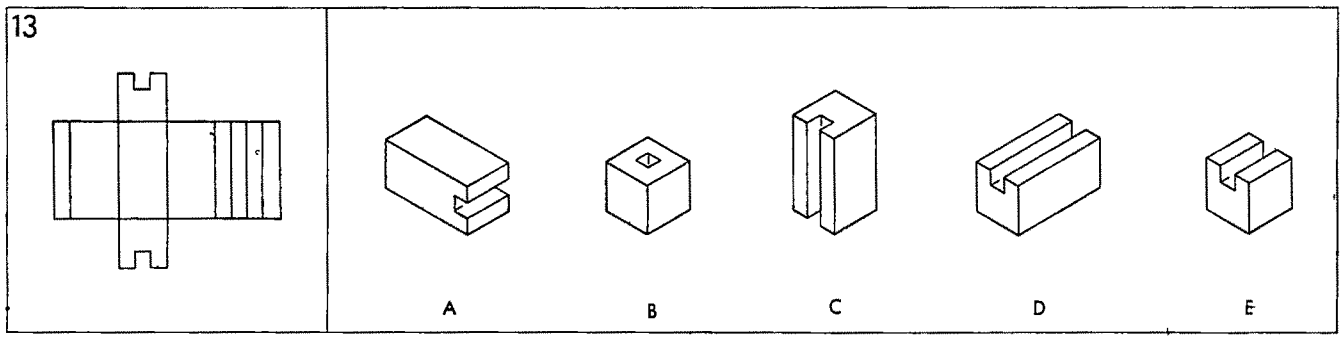
11



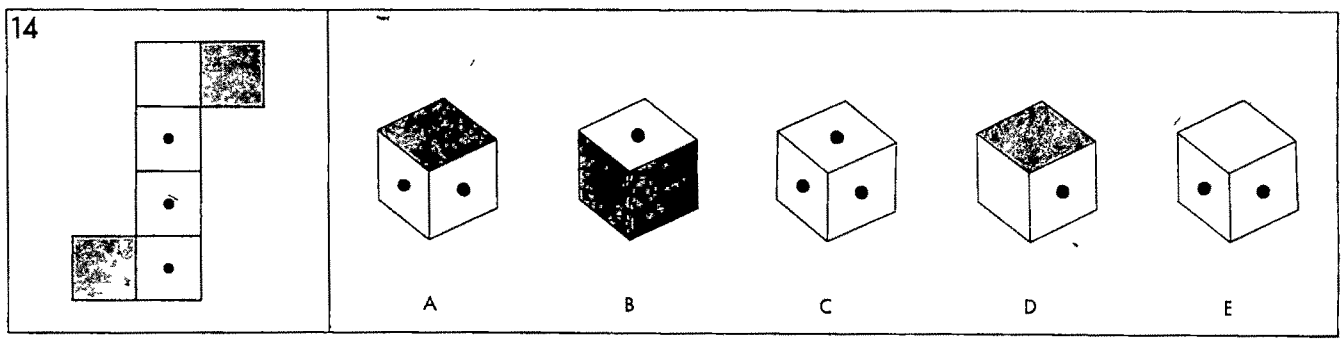
12



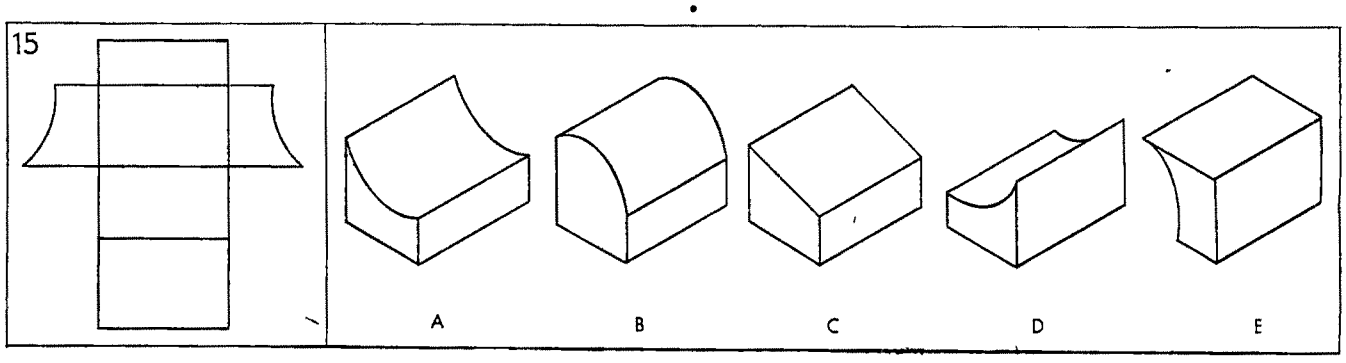
13



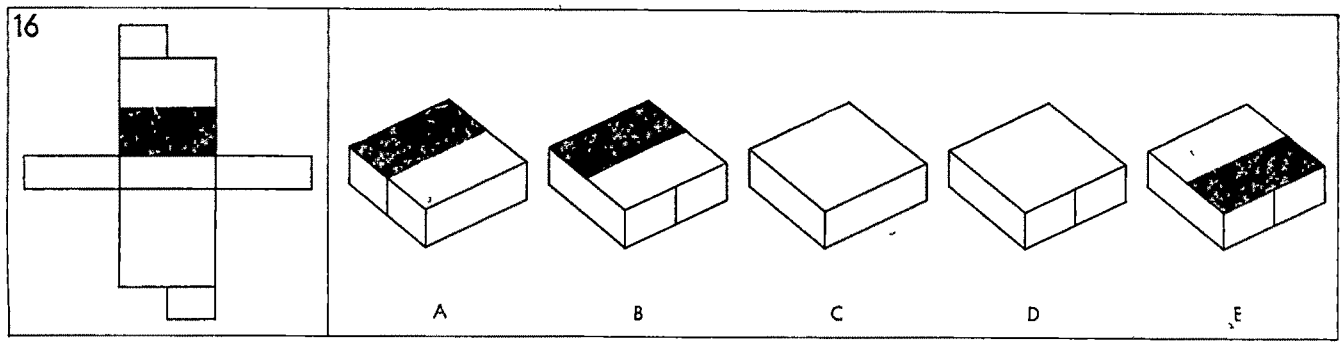
14



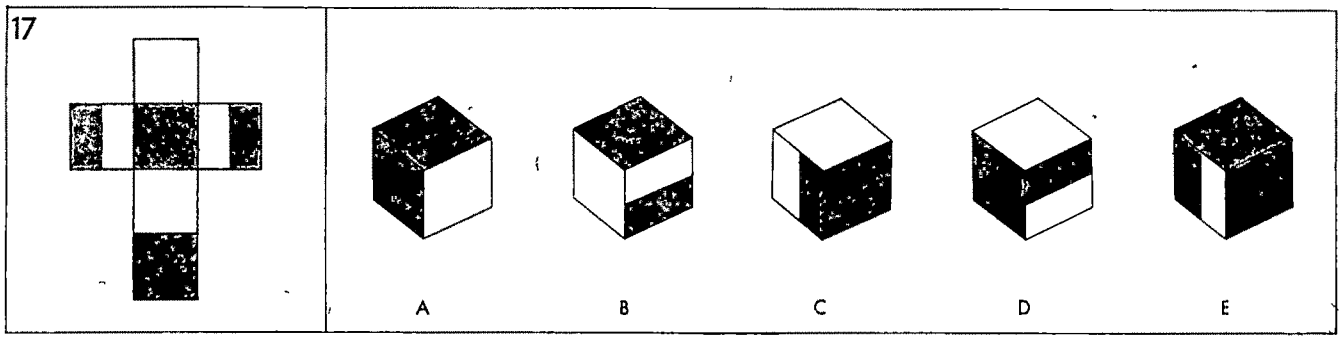
15



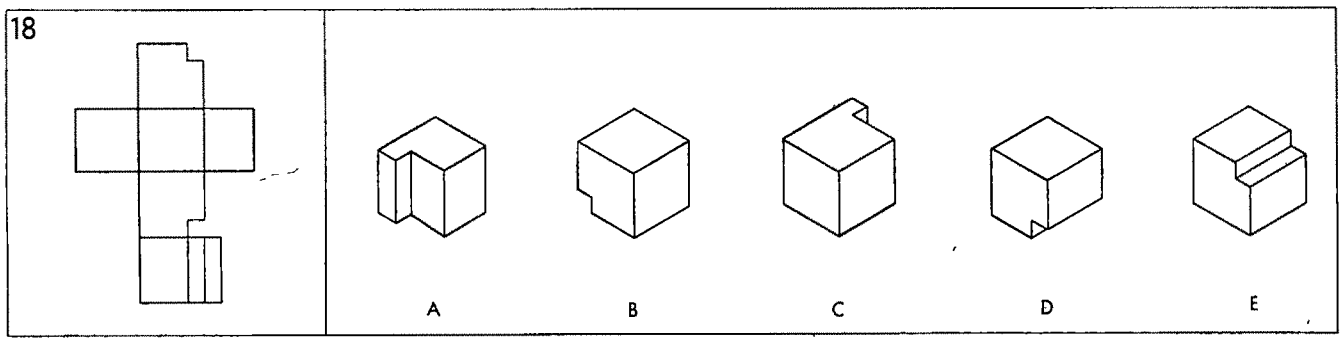
16



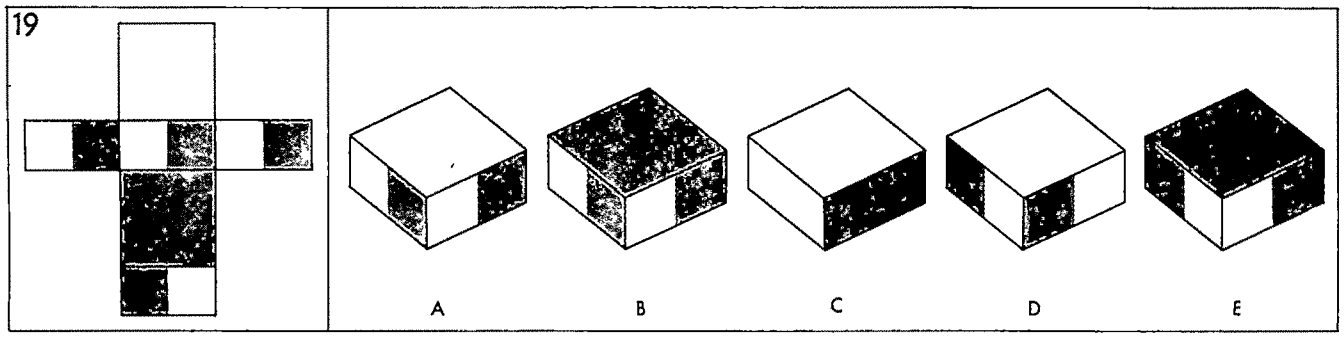
17



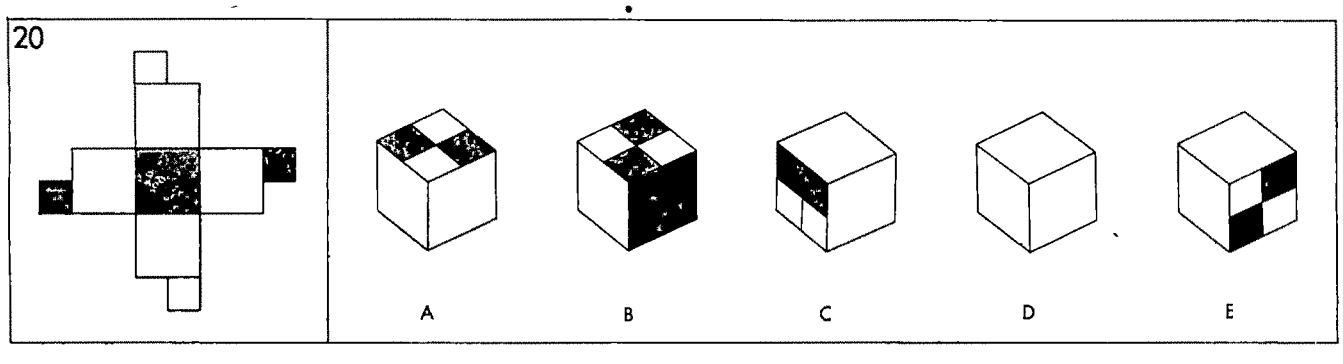
18



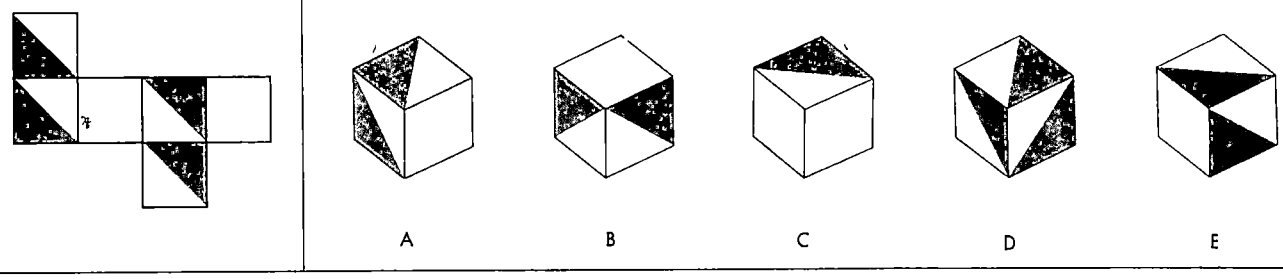
19



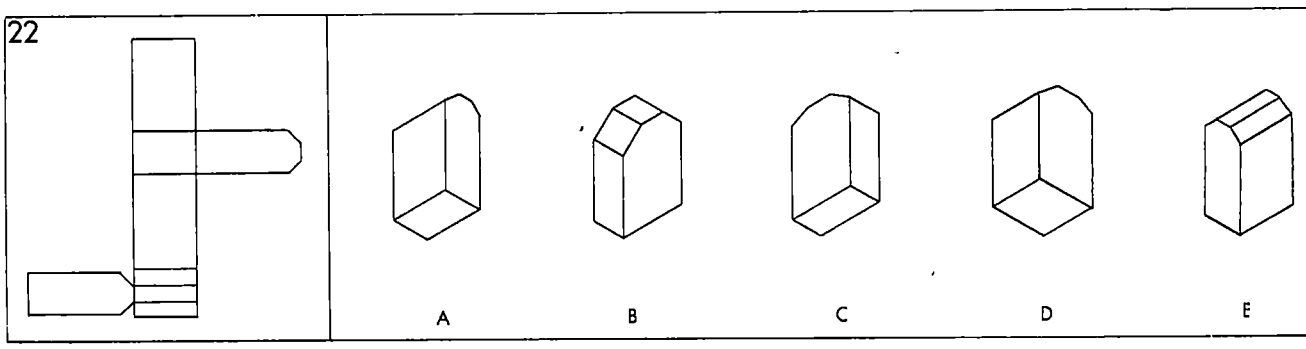
20



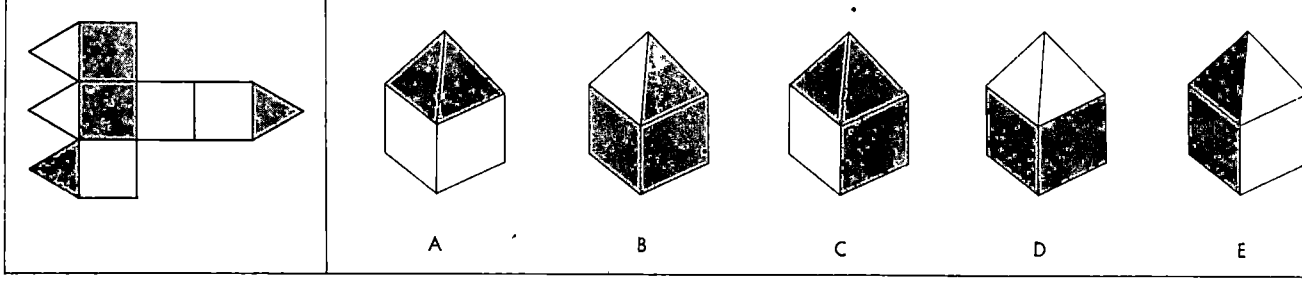
21



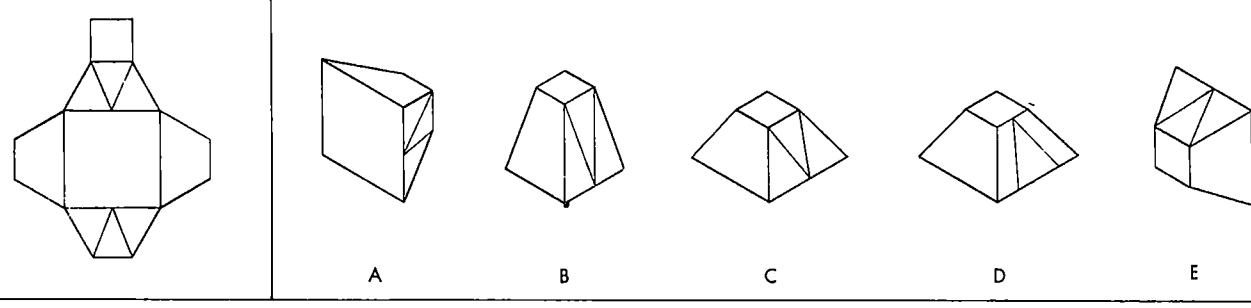
22



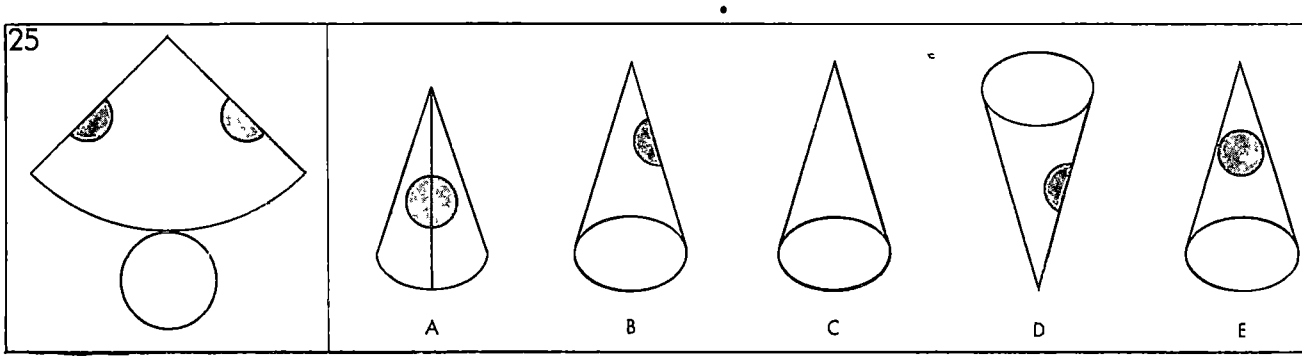
23



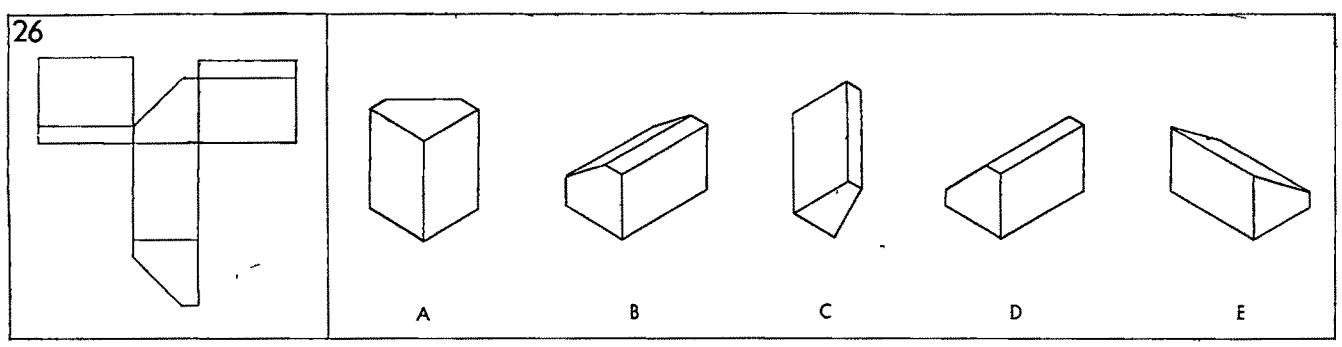
24



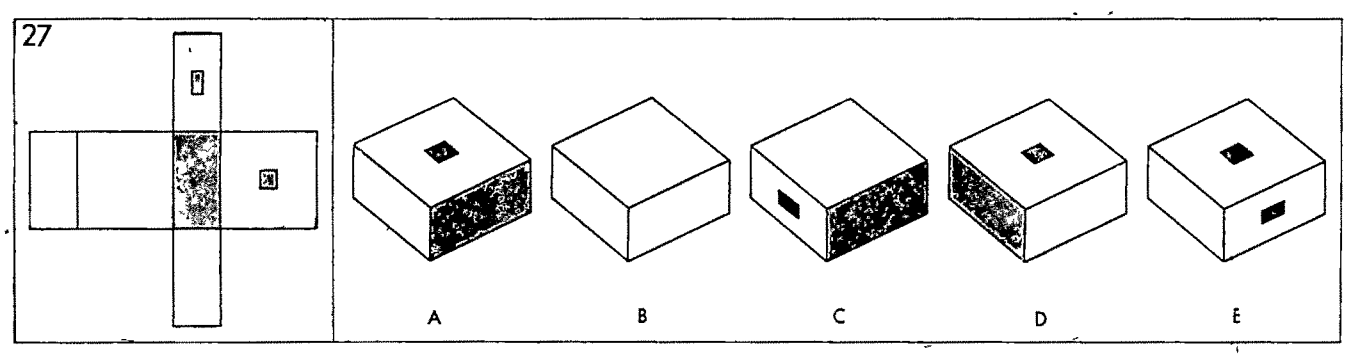
25



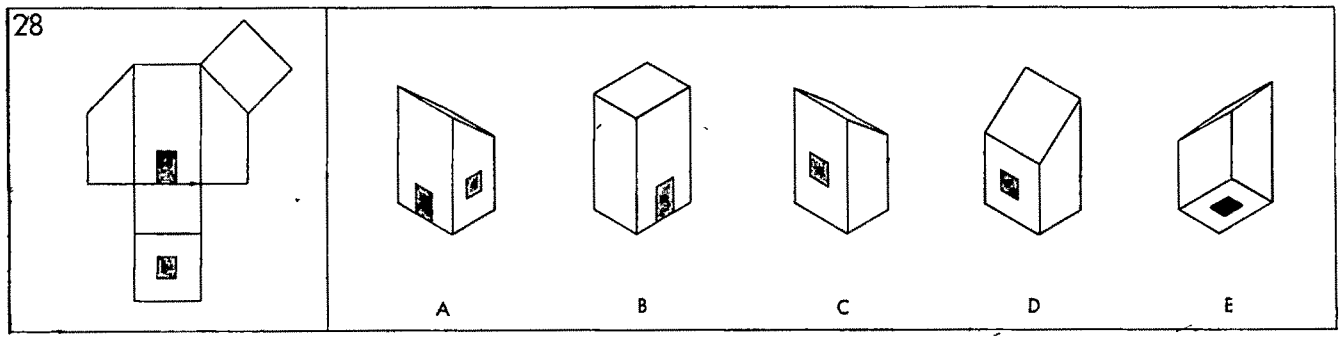
26



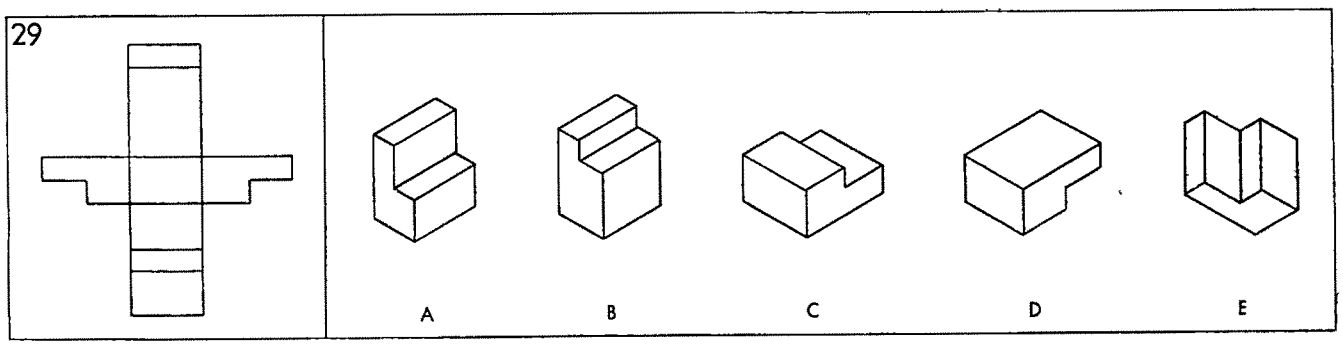
27



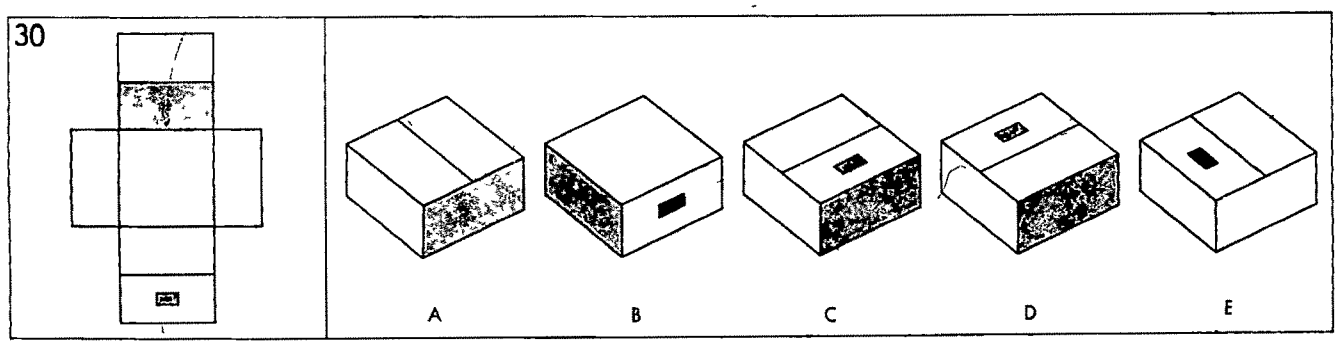
28



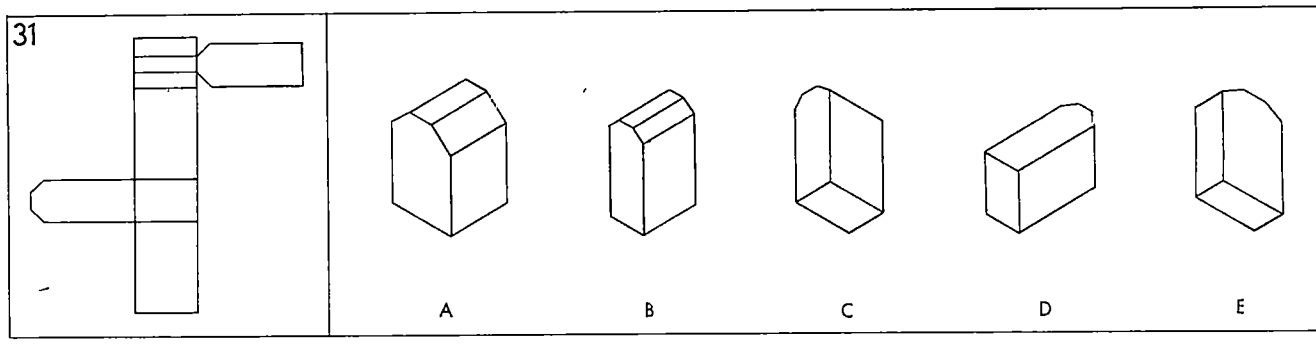
29



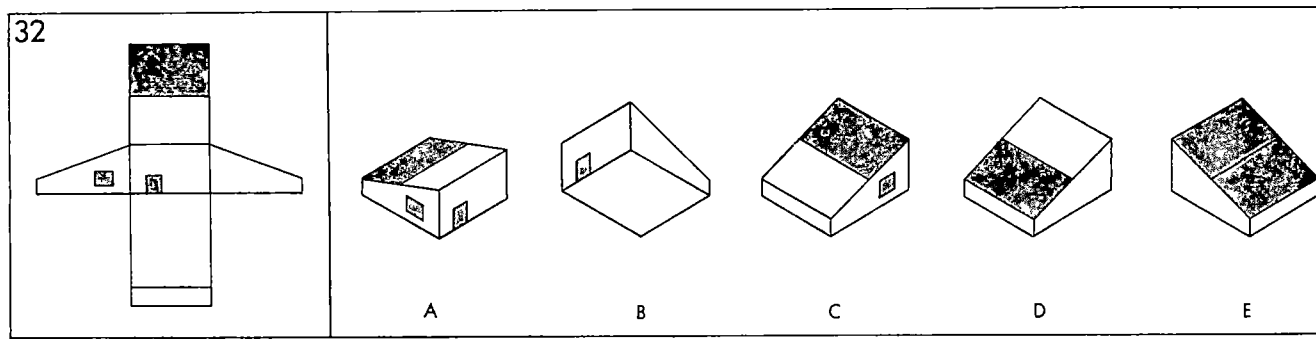
30



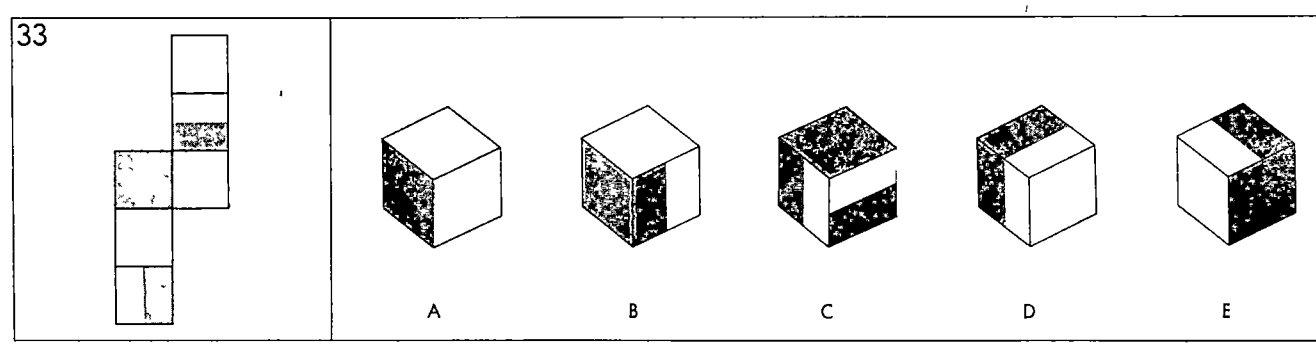
31



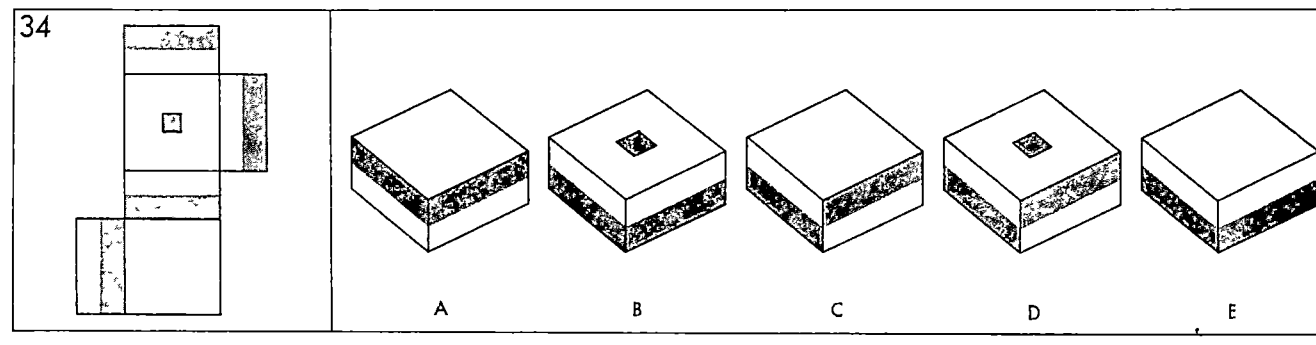
32



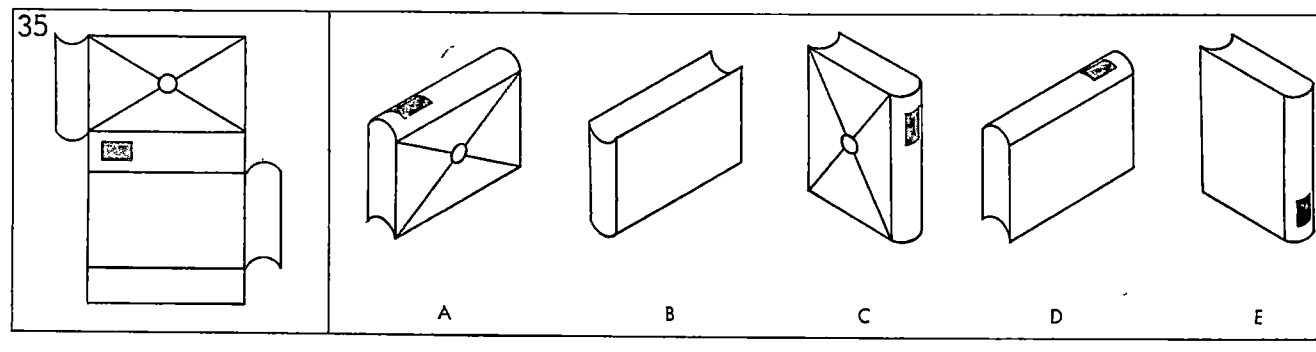
33



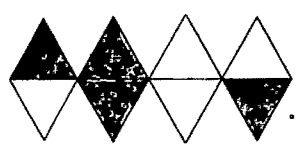
34

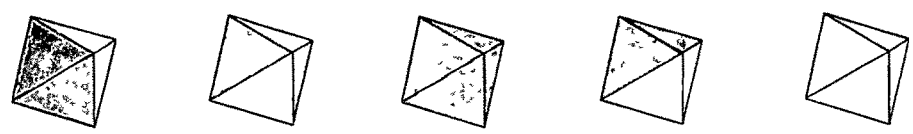


35



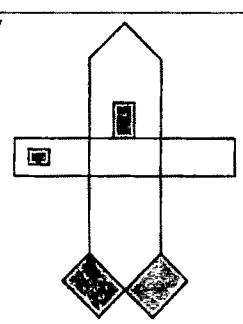
36

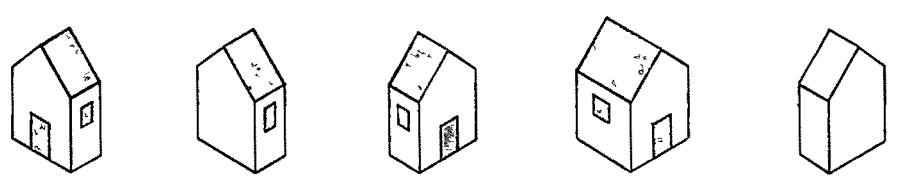




A B C D E

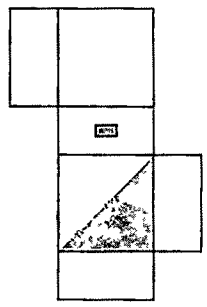
37

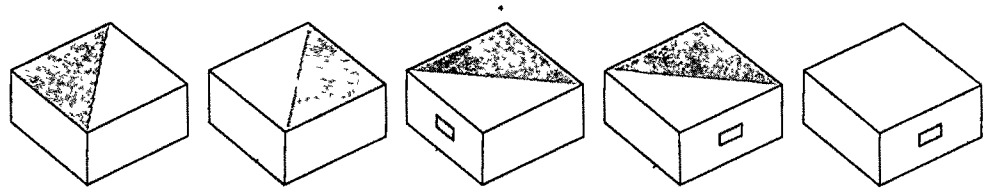




A B C D E

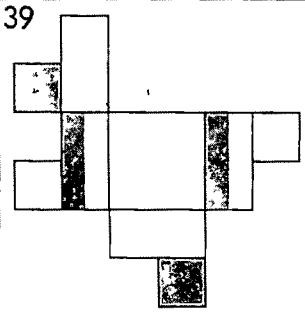
38

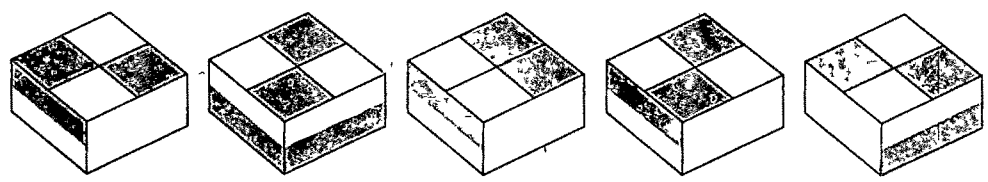




A B C D E

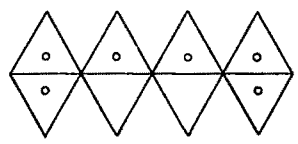
39

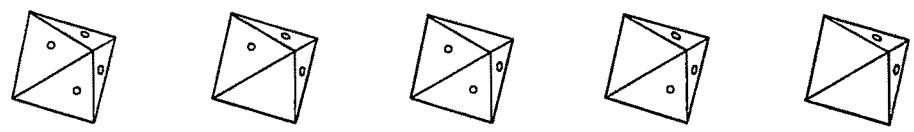




A B C D E

40





A B C D E

Appendix E 2

DIFFERENTIAL APTITUDE TESTS

SPACE RELATIONS

ANSWER SHEET

FORM ____

DO NOT WRITE IN THIS SPACE

RAW SCORE ____ PERCENTILE ____

NORMS USED ____

NAME ____ DATE ____

PLACE (SCHOOL) ____ CITY ____

AGE ____ YEARS ____ MONTHS SEX ____ MALE OR FEMALE CLASS ____

ERASE COMPLETELY ANY ANSWER YOU WISH TO CHANGE.

	A	B	C	D	E		A	B	C	D	E
1	○	○	○	○	○	21	○	○	○	○	○
2	○	○	○	○	○	22	○	○	○	○	○
3	○	○	○	○	○	23	○	○	○	○	○
4	○	○	○	○	○	24	○	○	○	○	○
5	○	○	○	○	○	25	○	○	○	○	○
6	○	○	○	○	○	26	○	○	○	○	○
7	○	○	○	○	○	27	○	○	○	○	○
8	○	○	○	○	○	28	○	○	○	○	○
9	○	○	○	○	○	29	○	○	○	○	○
10	○	○	○	○	○	30	○	○	○	○	○

BE SURE YOUR MARK IS INSIDE THE CIRCLE

	A	B	C	D	E		A	B	C	D	E
11	○	○	○	○	○	31	○	○	○	○	○
12	○	○	○	○	○	32	○	○	○	○	○
13	○	○	○	○	○	33	○	○	○	○	○
14	○	○	○	○	○	34	○	○	○	○	○
15	○	○	○	○	○	35	○	○	○	○	○
16	○	○	○	○	○	36	○	○	○	○	○
17	○	○	○	○	○	37	○	○	○	○	○
18	○	○	○	○	○	38	○	○	○	○	○
19	○	○	○	○	○	39	○	○	○	○	○
20	○	○	○	○	○	40	○	○	○	○	○

PUBLISHED & PRINTED IN INDIA BY
MANASAYAN, DELHI

Modified and Reproduced by Special Permission of the original publisher and copyright owner, THE PSYCHOLOGICAL CORPORATION, NEW YORK, NEW YORK. All rights reserved.